The Embodiment Abstraction: A Foundation Layer for Physical AI

A Position Paper

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Project page: `https://github.com/robotfoundationlayer`

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1 How to read this document

This is a **position paper**, not a peer-reviewable academic paper. It combines three kinds of content that an academic paper would normally separate:

1. **A specification proposal** (§§ 3–4, Appendix B). This is the technical core: the five constitutional principles, the three-layer specification, and the supermodularity property that motivates the bilateral structure.
2. **A network-effect and ecosystem analysis** (§ 5). This is illustrative and conceptual rather than predictive — the "Master Equation" of § 5.5 in particular has a free scaling coefficient and is explicitly an explanatory model rather than a calibrated forecast.
3. **A call for founding partners** (§ 9). This is a recruitment instrument for the seven-member founding cohort that will ratify spec v0.1.

1.1 Relationship to the published RFL Specification v0.1

A compact spec-only extract — academic register, no network-effect framing, no recruitment material — is **published as a companion artifact**: the **RFL Specification v0.1** at `https://github.com/robotfoundationlayer/rfl/blob/main/whitepaper/RFL_SPEC_v0.1_en.pdf` (62 pages). The two documents are designed to be read independently:

- **Read the spec extract** if your interest is the technical specification only — the contract, the invariants, the conformance regime, the reference algorithmic recipe. The extract carries the architectural commitments at academic register and is the artifact arXiv reviewers and downstream implementers are expected to focus on.
- **Read this position paper** if your interest extends to the network-effect framing, the Master Equation, the full governance design, the historical comparative arc (ARM/CUDA/USB at length), or the Founding Consortium recruitment frame. This document treats the spec as one component of a broader coordination programme and surrounds it with the strategic and institutional material the spec extract deliberately omits.

Concretely, content that this position paper develops in full and the spec extract only references:

- The Master Equation derivation (§ 5.5 here; the spec extract retains only the conditional supermodularity claim in its Appendix B).
- The nine reinforcement loops (§ 5.4 here; the spec extract does not treat the loop architecture).
- The full Governance design including Linux Foundation subsidiary structure (§ 7 here; the spec extract reduces this to the five constitutional commitments C1–C5 with operational mechanism for C1 and C4, and refers governance institutional framing to this document).
- The Call to Action and the Founding Consortium recruitment proposal (§ 9 here; the spec extract neutralizes this to "implementer cohort" language).
- The Crémer-McLean structural parallel and the empirical-evidence discussion for bilateral compounding (Appendix D here; the spec extract removed these per external review feedback that they overstated what the structural argument establishes).

Cross-references in this document of the form "§ X of the spec extract" point to the corresponding section of the published RFL Specification v0.1; cross-references in the spec extract of the form "described in the companion position paper" point to the corresponding section of this document. The two documents share Abstract content, Keywords, References, and Appendix A (TactileManifold formal specification) by construction; everything else is independent.

Abstract

The physical-AI ecosystem of 2026 faces a coordination problem. Approximately nine credible Vision-Language-Action (VLA) foundation models and approximately thirty distinguishable robotic embodiments yield a combinatorial integration burden plausibly large enough to constrain industrial deployment (no published benchmark currently exists; the paper commits to one in v0.2). Adjacent efforts — ROS 2, Open X-Embodiment, LeRobot, the single-model VLAs themselves, and NVIDIA's Isaac and GR00T stack — each address fragments of the problem but none occupies the specific structural position the situation appears to demand: a cross-vendor, semantically-typed, production-grade, neutrally-governed abstraction layer between intelligence and embodiment. We propose the Robot Foundation Layer (RFL), a three-layer specification consisting of (i) a Skill ISA of approximately fifty primitives with a compositional algebra, (ii) a canonical embodiment-agnostic action representation with a deterministic retargeting interface contract, and (iii) a Driver Interface compatible with ROS 2, all governed by five first principles: embodiment-agnostic, compositional, verifiable, provider-neutral, and forward-compatible. The retargeting contract is the load-bearing technical commitment of the specification; this paper defines the contract and its five binding invariants but leaves the implementing algorithm to the v1.0 reference implementation. We sketch an Appendix B argument that bilateral retargeting quality is plausibly supermodular under standard assumptions, suggesting a structural compounding property whose empirical magnitude is left to deployment. The specification draft, reference-implementation skeleton, and conformance test suite scaffold are released under Apache License 2.0 at the project page; comments and structural critique are solicited through the v0.1 review period.

Keywords

physical AI, vision-language-action models, robotic manipulation, foundation models, abstraction layers, two-sided markets, network effects, standards, ARM ISA, open governance

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1.2 Table of Contents

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2 § 1. The Embodiment Crisis

The capability gap between what AI models can describe and what robotic systems can perform has, in the past three years, inverted. As recently as 2023, the binding constraint on physical AI was the model — robots could execute well-defined motions but could not generalize across tasks because the underlying intelligence was too brittle. By the second half of 2025 this constraint had moved. Vision-Language-Action (VLA) foundation models — $\pi 0$ from Physical Intelligence, RT-2 and Gemini Robotics from Google DeepMind, Helix from Figure, GR00T N1 from NVIDIA, OpenVLA from Stanford, and a growing cohort of open-source successors — now generalize across object categories, manipulation primitives, and natural-language instructions in ways that, a decade ago, would have been classified as artificial general intelligence in the manipulation domain.

The binding constraint has moved to the boundary between these models and the physical bodies they are supposed to inhabit. Every VLA must be re-integrated, by hand, against every embodiment it is to control. Every embodiment vendor must repeat this integration with every VLA it wishes to support. The cost of this combinatorial work has, in 2026, become the dominant drag on the field — larger than model training cost, larger than hardware unit cost, and larger than data acquisition cost. This section quantifies that drag, decomposes its sources, and argues that the present window — roughly the eighteen months bracketing 2026 — is the inflection at which the industry must coordinate on a shared abstraction or accept a generation of vertically integrated lock-in.

2.1 1.1 The N $\times$ M Explosion

Two enumerations frame the problem.

On the **model side**, approximately nine VLA foundation model lines are credible candidates for industrial deployment in 2026, by which we mean: they have published checkpoints or commercial APIs, they have demonstrated multi-task generalization in third-party evaluations, and they are actively maintained by an organization with twelve-month forward visibility. The set, in alphabetical order: Figure's Helix; Google DeepMind's Gemini Robotics (current released line, e.g., Robotics-ER 1.6 announced April 2026); Hugging Face's SmolVLA and VLA-JEPA model lines; NVIDIA's Isaac GR00T family (current line, e.g., N1.7); Physical Intelligence's $\pi 0$ family ($\pi 0$ released October 2024 and open-sourced February 2025, $\pi 0.5$ released April 2025, $\pi 0.7$ released April 2026); Stanford / UC

Berkeley / Toyota Research Institute's OpenVLA; Tsinghua's RDT-1B (ICLR 2025); and the Octo family from UC Berkeley. This count is conservative — it excludes academic checkpoints without industrial maintenance commitments and several proprietary models known to be in development at OpenAI, Microsoft, Apple, and others but not publicly disclosed.

On the **embodiment side**, the count is larger and growing faster. Restricting to manipulation-capable end effectors and arms with commercial or near-commercial availability as of mid-2026, the set includes: humanoid platforms from Tesla (Optimus, with Optimus 3 production beginning summer 2026), Figure (02 and 03, with Figure 03 fleet commercially deployed at BMW), 1X (NEO, consumer preorders open), Sanctuary AI (Phoenix), Apptronik (Apollo, in active customer trials), Agility Robotics (Digit), Boston Dynamics (electric Atlas, serial production from March 2026), Unitree (G1 at approximately $16k, H1, and H2), and a cohort of Chinese entrants from Xiaomi, ByteDance, UBTech, and others; multi-finger hands from Inspire Robotics (RH8D, RH8R), Wonik (Allegro Hand), CMU (LEAP Hand, open source), Shadow Robot Company (Dexterous Hand v4), Schunk (SVH), Bridgestone (TETOTE pneumatic series), and roughly twenty research-grade hands maintained by academic labs; and conventional collaborative arms from Universal Robots, Franka Emika, Kuka, ABB, UFactory (xArm), Doosan, and Yaskawa. Conservatively de-duplicated, this yields approximately **thirty distinguishable embodiments** in active 2026 deployment, with credible projections of fifty by end-of-2027 as the humanoid cohort matures.

The combinatorial implication is direct. Nine VLAs and thirty embodiments generate $9 \times 30 = 270$ candidate integration pairs. Even if we accept that not every pair is commercially meaningful — Tesla Optimus paired with Tsinghua RDT-1B, for example, has no obvious go-to-market — the realistically valuable subset still exceeds one hundred pairs. As the humanoid wave matures and the VLA cohort expands toward the dozen-plus models that the major foundation-model laboratories will field by 2027, the realistically valuable subset crosses two hundred. **No actor in the field is equipped to maintain anything approaching this number of integrations.**

The current response, observable in 2026, is vertical lock-in: each VLA developer prioritizes a small number of embodiments (Physical Intelligence with a handful of arms and hands, Helix exclusively with Figure's humanoid, GR00T N1 with NVIDIA-blessed reference robots), and each embodiment developer prioritizes a small number of VLAs. The market thereby fragments into eight or ten vertically integrated stacks, each of which captures a fraction of the latent value because no stack achieves the cross-pollination that drove every previous wave of platform progress in computing.

2.2 1.2 The Integration Tax

What does a single (VLA, embodiment) integration actually cost? The honest answer is that the field lacks rigorous public benchmarks, but converging private and academic estimates allow a defensible range.

A single non-trivial integration — defined as bringing a VLA from supporting one embodiment to supporting an additional one, at production-grade reliability — consumes between **two and six engineer-months** of senior robotics-AI talent. The lower bound applies when the new embodiment is structurally similar to one the VLA already supports (e.g., adding a second 7-DOF arm of the same kinematic class). The upper bound applies when the embodiment introduces novel actuation modality (pneumatic versus tendon-driven), novel proprioceptive sensors (visuotactile versus force-only), or novel kinematic structure (a six-finger McKibben hand instead of a four-finger tendon-driven hand). A pneumatic six-finger hand — structurally unlike anything in the standard learning-research toolkit — sits squarely at the upper bound of this range.

Senior robotics-AI engineers, in 2026, command fully-loaded compensation in the high tens of thousands of dollars per engineer-month in the dominant US labor market — consistent with senior-IC bands at leading AI laboratories — with a meaningful step-down for similar talent in Europe and Asia but offset by reduced availability. Per-pair direct labor cost therefore ranges, plausibly, from the low tens of thousands of dollars at the lower bound to the low hundreds of thousands at the upper bound. To this we must add three indirect costs that are nontrivial.

First, **hardware procurement and access**: each new embodiment requires acquiring at least one unit, instrumenting it for data collection, and maintaining it through the integration window. Hardware acquisition cost ranges from a few thousand dollars for low-cost research-grade hands to hundreds of thousands for humanoid platforms; the amortized per-integration cost of hardware access typically lands in the low tens to low hundreds of thousands of dollars.

Second, **data collection and labeling**: cross-embodiment generalization in VLA training requires substantial demonstration data from the target embodiment. Teleoperation rigs, expert demonstrators, and curation pipelines push another contribution in the tens of thousands to low hundreds of thousands of dollars per integration, depending on task complexity.

Third, **certification and safety qualification**: any embodiment intended for industrial deployment must pass ISO 10218 or ISO 13482 safety assessment when paired with novel control software. This certification is rarely budgeted in research-grade integrations but becomes a line item in the low to mid hundreds of thousands of dollars when commercial deployment is the target.

The total per-pair integration cost therefore plausibly ranges from the **low hundreds of thousands of dollars at the lower bound (research integration of a similar embodiment) to the high hundreds of thousands at the upper bound (commercial integration of a structurally novel embodiment)**, with a small number of disclosed integrations exceeding a million dollars when novel actuation modalities are involved. We adopt **$300,000 as the median estimate** for non-trivial commercial integrations as of mid-2026, based on the authors' industry experience; it is intended as a working figure rather than a precise measurement.

2.3 1.3 The Fragmentation Tax

Aggregating across the field, the integration tax is large enough to be macroscopic. If the realistically valuable subset of integrations in 2026 is one hundred pairs, and the median per-pair cost is approximately $300,000, the total annual integration spend across the global physical-AI sector approaches **the low tens of millions of dollars per year measured at the marginal level**. This is a conservative figure because it counts only first-time integrations; the ongoing maintenance burden

— keeping a working integration current as the VLA evolves and the embodiment firmware is updated — typically runs at twenty to thirty percent of the original integration cost per year, adding another contribution in the high single-digit millions annually.

The aggregate figure understates the true drag, however, because the more important cost is not what gets paid but what does not get attempted. A research lab evaluating whether to test its new VLA on a novel six-finger hand must trade six months of senior engineering time against the project's other commitments; in most cases, the test does not happen. A startup considering a new robot form factor must accept that VLA support will lag the hardware by twelve to eighteen months. A potential industrial customer evaluating whether to deploy a robotic solution must price in not only the hardware and software they buy but the integration risk of pairing them. **The opportunity cost of attempts not made dwarfs the direct cost of attempts made.**

We can decompose the fragmentation tax into four components:

1. **Direct integration spend**: the low-tens-of-millions annual figure above, paid by the firms doing the integrations.
2. **Maintenance overhead**: the 20–30% annual recurrence on prior integrations.
3. **Capability suppression**: the value of integrations not attempted because the cost is prohibitive. This is the largest component but the hardest to estimate directly. A defensible proxy is the gap between observed and counterfactual market growth in segments where integration cost is binding; this gap is plausibly on the order of high hundreds of millions to low billions of dollars in annual commercial value, though it resists precise empirical measurement and we present no point estimate.
4. **Lock-in penalty**: the long-run cost of customers being trapped in vertically integrated stacks they would have left if cross-vendor migration were lower friction. Historical analogs from CPU and operating-system markets suggest this penalty compounds at roughly the rate of the underlying market growth, so it grows even when no new integrations are attempted.

Taken together, the fragmentation tax — direct, maintenance, suppression, and lock-in — is plausibly on the order of **low single-digit billions in 2026** and is on a trajectory toward the low tens of billions by 2030 as the humanoid wave and the VLA cohort both expand. Both figures are dominated by the capability-suppression component and therefore inherit its uncertainty. This is not a research curiosity. It is the most consequential structural drag on the physical-AI industry's progress.

2.4 1.4 Why 2026 Is the Inflection Point

The combinatorial pressures described above have existed in some form since the field's inception. What makes 2026 distinctive — and what makes coordination on a shared abstraction layer urgent in this specific window — is the simultaneous occurrence of three inflections that, individually, would each be noteworthy.

Inflection one: VLA models cross the threshold of cross-embodiment generality. Through 2024, the dominant assumption in the field was that a model trained on one embodiment would transfer poorly to another and that cross-embodiment generalization would require fundamentally new architectures. The Open X-Embodiment results published by Google and thirty-three partner institutions in 2023, followed by π0's multi-embodiment training in 2024 and Gemini Robotics's transfer results in 2025, falsified this assumption. By 2026 the question is no longer whether a single VLA can support many embodiments but rather *how many embodiments will be available to support*. The model side is suddenly hungrier than the embodiment side can feed.

Inflection two: embodiments diversify faster than they consolidate. The humanoid wave has, against many expert predictions, not produced a single dominant form factor. Tesla Optimus's five-finger tendon design, Figure 03's twenty-four DOF hand, Sanctuary's hydraulic actuation, Bridgestone TETOTE's six-finger pneumatic structure, and the emerging Chinese cohort all represent distinct kinematic and actuation choices, each of which has a credible industrial sponsor and a credible commercial trajectory. The expected consolidation toward a single anthropomorphic form has not occurred and shows no sign of occurring on a timeline shorter than a decade. The embodiment side is therefore proliferating, not concentrating, at exactly the moment the model side is demanding it.

Inflection three: capital is committed at unprecedented scale, but coordination capital is missing. Through 2024 and 2025 the physical-AI sector raised disclosed venture financing in the tens of billions of dollars, with humanoid robotics accounting for a substantial fraction — by independent aggregator estimates, on the order of half the total. This capital has flowed almost entirely into vertical integration: into VLA model training, into specific robot designs, into proprietary data collection. None of it, to a useful approximation, has flowed into the horizontal coordination layer that would multiply the value of the vertical investments. The field has solved the funding problem for individual stacks and produced a coordination problem in its place.

These three inflections compound. A field with stagnant model generality, consolidating embodiments, or limited capital would not need an abstraction layer urgently — bilateral integration would scale acceptably. A field with all three growing simultaneously, as in 2026, requires a layer that arbitrates the combinatorial growth or it will fragment into stacks that capture, in aggregate, a fraction of the available value.

The historical analog is instructive. The personal-computer industry of the early 1980s faced a structurally similar moment: software was becoming general enough to demand cross-hardware portability, hardware was diversifying rather than consolidating, and capital was abundant. The coordination layer that emerged — the IBM PC architecture, eventually inherited by the x86 ISA and the BIOS conventions that became the de facto standard — captured immense value precisely because it appeared in the brief window when the alternative (per-vendor verticalization) was just barely tolerable. A coordination layer proposed five years earlier would have lacked the demand pull; five years later would have arrived after the lock-in had ossified. The window was eighteen months, perhaps two years, and the right answer in that window shaped the industry for the next four decades.

2.5 1.5 The Crisis Stated, the Question Posed

The diagnosis is therefore precise. The physical-AI industry of 2026 is converging on a coordination failure: model generality is high, embodiment diversity is high, capital is high, and the integration interface between models and bodies is bespoke and bilateral. The fragmentation tax is plausibly on the order of low single-digit billions annually — a figure dominated by the capability-suppression component and inheriting its uncertainty (§ 1.4) — and accelerating. The opportunity cost of integrations not attempted dwarfs the direct cost of integrations made. Vertical integration has emerged as the rational firm-level response to the absence of a coordination layer, and the resulting balkanization is locking in the precise structure that long-run progress in physical AI cannot afford.

The remainder of this paper argues that the appropriate response is the construction of a neutral, well-governed abstraction layer — the Robot Foundation Layer (RFL) — that occupies the same structural position in physical AI that ARM occupies in silicon, that USB occupies in peripheral interconnects, and that POSIX occupies in operating systems. § 2 reviews why existing efforts (LeRobot, Open X-Embodiment, ROS 2, NVIDIA Isaac, and others) have each failed to occupy this position despite addressing adjacent concerns. § 3 develops the first principles from which the RFL design follows. § 4 presents the specification at version 0.1. § 5 formalizes the network-effect structure on which RFL's long-term defensibility rests. § 6 lays out the implementation roadmap; § 7 describes the Foundation that will govern the standard; and §§ 8–9 compare RFL to its historical analogs and invite the founding consortium.

The window is open. It will not remain so indefinitely.

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3 § 2. Why Existing Approaches Fail to Occupy the Coordination Layer

The coordination crisis described in § 1 is not invisible to the field. Substantial engineering effort over the past five years has gone into systems that address adjacent fragments of the problem, and several have accumulated real momentum. Yet none has converged on the specific structural position that the crisis demands: a cross-vendor, semantically-typed, production-grade abstraction layer governed neutrally between Vision-Language-Action (VLA) models and robotic embodiments. This section examines the five most credible candidates — ROS 2, Open X-Embodiment, LeRobot, the single-model VLAs themselves, and NVIDIA's Isaac/GR00T stack — and argues that each is structurally precluded from occupying the position even on its natural roadmap. The argument is not that these projects are wrong or unimportant; they are correct and important within the problems they were built to solve. The argument is that none of those problems is the same as RFL's problem.

3.1 2.1 ROS 2: A Communication Layer Without Semantics

The Robot Operating System, in its second-generation form (ROS 2, built on the Data Distribution Service standard), is the closest thing the robotics field has to a universally adopted convention. Its action-server abstraction, its message-type system, and its publish-subscribe transport are deployed in thousands of research and commercial robots. If any existing system could expand upward into the coordination role RFL would occupy, ROS 2 is the prima facie candidate.

It cannot. ROS 2's design center is the low-level transport of typed messages between processes — "send a `geometry_msgs/Twist` to topic `/cmd_vel`" — and its action-server abstraction defines the *shape* of long-running operations (goal, feedback, result) but not their *meaning*. There is no canonical ROS 2 definition of what it means for a robot to "grasp a cable end and insert it into a connector"; there is only a convention for how such an action's interface might be structured if defined. The semantic gap between "I can transport bytes reliably" and "I can convey the meaning of *grasp* across two different hands so that a single high-level model can drive both" is precisely the gap RFL must close.

Expanding ROS 2 to close this gap is not a roadmap question; it is an identity question. The current ROS 2 contract — backward-compatible, transport-focused, runtime-agnostic — would have to be broken to introduce action semantics, and breaking it would alienate the deployed base on which its credibility rests. The right move for ROS 2 is to remain the transport layer beneath RFL, not to attempt to subsume it. § 4 of this paper makes the integration with ROS 2 explicit: RFL's Driver Interface is designed to expose itself as a set of ROS 2 action servers, so that the existing transport investment is preserved.

3.2 2.2 Open X-Embodiment: A Dataset Without a Standard, a Stewardship Without Neutrality

Open X-Embodiment, the Google-led initiative joined by thirty-three partner institutions in 2023, has done more than any other single effort to establish the empirical foundation for cross-embodiment generalization. The pooled dataset — over a million teleoperated trajectories spanning twenty-two embodiments — supplied the training corpus on which RT-2, $\pi 0$, and subsequent VLAs demonstrated their cross-platform transfer results. The intellectual contribution is substantial.

The structural contribution, however, stops at the dataset boundary. Open X-Embodiment defines a *data format* (largely the RLDS format from Google's prior data work) but not a runtime standard. It tells researchers how to label demonstration trajectories from a new embodiment; it does not tell a production deployment how to route a high-level instruction from a VLA to that embodiment at runtime. A VLA trained on Open X-Embodiment data still requires bespoke integration to control a new embodiment in deployment — the dataset standard does not bridge to a control standard. The same boundary applies to demonstration-capture interfaces such as the Universal Manipulation Interface (UMI), which standardize how manipulation demonstrations are collected in the wild, including the hand-held gripper hardware and the policy-learning pipeline, but not how a model's intent is retargeted to and verified across heterogeneous embodiments at runtime. These are complementary upstream infrastructure for *training* the planner, not the cross-vendor runtime contract RFL occupies.

The second blocker is stewardship. Open X-Embodiment is housed at Google, and Google maintains its own VLA model line (Gemini Robotics) that competes commercially with other VLA providers. This does not impugn the integrity of the academic collaboration, but it does foreclose the trajectory by which Open X-Embodiment could evolve into the neutral coordination layer. The OEM that adopts an abstraction standard housed at a hyperscaler with its own VLA product is making a different commercial bet than the OEM that adopts a standard housed at a neutral foundation. Cisco and 3Com were never going to adopt a networking standard owned by IBM; the same dynamic applies here.

3.3 2.3 LeRobot: Strong Momentum, Wrong Center of Gravity

LeRobot, the open-source robotics framework that Hugging Face introduced in 2024 and has aggressively scaled through 2025–2026, is the candidate most often cited in conversations about coordination-layer convergence. It has the right cultural disposition (open source, community-driven), the right developer-experience instincts inherited from the Hugging Face transformers ecosystem, and the most rapidly accumulating contributor base of any project in the field. If RFL is to encounter a serious competitor for the structural position it seeks, the realistic candidate is LeRobot.

The reason LeRobot does not yet — and structurally may not — occupy the RFL position lies in two divergent design pressures. First, LeRobot's center of gravity is the ML-researcher workflow: train a policy, evaluate it on a benchmark, share the checkpoint. This is the right design center for accelerating research, and LeRobot has done it superbly. It is not the right design center for production deployment in industrial environments, which require real-time safety guarantees, certifiability against ISO 10218/13482, deterministic latency, and integration with existing PLC and field-bus infrastructure. The features that make LeRobot a great research framework — Python-first orientation, flexibility, rapid iteration — work against the deterministic, statically analyzable, safety-certifiable substrate that industrial deployment demands.

Second, Hugging Face's commercial trajectory tends toward vertical platform consolidation rather than standards stewardship. The Hugging Face Hub is itself a platform with proprietary monetization (Spaces, Inference Endpoints, Enterprise tier), and the natural commercial path for LeRobot is to become the gateway to Hugging Face's robotics services. This is a legitimate business model, but it is structurally incompatible with the neutral-foundation governance that the coordination layer requires. An OEM that builds on LeRobot is building on infrastructure whose operator competes commercially with adjacent layers — exactly the trap that ARM avoided by separating itself from Acorn.

A possible counter is that Hugging Face could spin LeRobot out into a neutral foundation analogous to the Linux Foundation. We treat this as the most credible scenario in which the RFL position is contested rather than empty, and § 9's call to action explicitly invites collaboration with LeRobot stewards under such a structure. As of mid-2026 no such restructuring has been signaled.

3.4 2.4 Single-Model VLAs: Applications That Cannot Be Platforms

A more provocative claim than any of the above is that the leading VLA models themselves — $\pi 0$ from Physical Intelligence, OpenVLA from Stanford, GR00T N1 from NVIDIA — cannot become the coordination layer regardless of how technically capable they grow. The reasoning is structural, not technical, and it applies even to the most generous interpretation of any of these models' future capability.

A VLA is an application: an artifact that takes perception and instruction in and produces actions out. It competes, in the market, with other applications that do the same thing. Two VLAs competing for the same customer's deployment cannot simultaneously serve as the neutral substrate on which that customer's robot is built. Physical Intelligence's decision to open-source the $\pi 0$ weights does not change this; the weights are now free, but the integration interface — what counts as a valid $\pi 0$ input, what semantics its action outputs carry, how it expects embodiment metadata — remains controlled by Physical Intelligence and follows Physical Intelligence's commercial roadmap. An OEM that builds its embodiment-side software against $\pi 0$'s specific interface is taking a bet on Physical Intelligence, not on a standard.

The dynamic is identical to the early 1990s in operating systems: Sun Microsystems' Solaris was excellent, IBM's AIX was excellent, HP-UX was excellent, but none of them could serve as the substrate for a multi-vendor application ecosystem because each was a commercial product of a vendor that competed for the same customers. The substrate that did emerge — POSIX as a specification, eventually Linux as an implementation — was created precisely because no incumbent could occupy the structural position. The physical-AI field is at the same juncture with respect to VLAs.

The implication is not that $\pi 0$, OpenVLA, and GR00T are unimportant; they are the applications that will run on top of RFL. The implication is that none of them, on its present trajectory, can become RFL.

3.5 2.5 NVIDIA Isaac and GR00T: The CUDA-Adjacent Trap

NVIDIA's Isaac platform, paired with the GR00T N1 humanoid foundation model and the Cosmos world model, is the most technically capable adjacent stack on offer in 2026. It bundles simulation (Isaac Sim, Isaac Lab), perception, motion planning, foundation models, and deployment tooling into the most integrated end-to-end physical-AI offering currently available. By raw capability it is, plausibly, the strongest existing candidate.

By neutrality it is the weakest. Isaac's value proposition is inseparable from CUDA: the simulation runs on NVIDIA GPUs, the foundation models train on NVIDIA hardware, the inference deploys on NVIDIA edge devices (Jetson, Thor). This is not a deficiency in the engineering; it is the engineering. NVIDIA built Isaac to sell NVIDIA silicon, and the architecture reflects that goal at every layer.

The result is that any embodiment vendor or VLA provider that builds against Isaac is, structurally, accepting a CUDA dependency in perpetuity.

A coordination layer cannot be CUDA-dependent because half the relevant ecosystem is structurally incentivized to escape CUDA: Apple's Neural Engine, Google's TPU, Tesla's Dojo, Qualcomm's Hexagon, and an emerging cohort of ARM-based and RISC-V-based edge accelerators all need a path to physical-AI deployment that does not route through NVIDIA's tooling. Even within NVIDIA's customer base, the AI-chip market's recent volatility has made customers cautious about deepening any single-vendor dependency. An abstraction layer whose adoption signals long-term lock-in to NVIDIA is, for half its potential adopters, worse than no abstraction layer at all.

The case is parallel to OpenGL versus Direct3D in the late 1990s: Direct3D was technically superior in many releases, but its Microsoft dependency drove the cross-platform graphics market to consolidate around the neutral alternative. RFL's relationship to Isaac is the same: cooperative where possible (RFL implementations should run on Isaac Sim as one of several supported simulators), structurally distinct in governance.

3.6 2.6 The White Space, Mapped

Set the five existing efforts against the coordination layer's requirements and the gap becomes geometrically visible.

Requirement	ROS 2	Open X-E	LeRobot	Single-model VLAs	NVIDIA Isaac	**RFL**
Cross-vendor VLA support	n/a	n/a	partial	by definition no	partial	✓
Cross-vendor embodiment support	partial	partial	partial	partial	partial	✓
Semantically-typed actions (not just messages)	×	×	partial	proprietary	proprietary	✓
Production-grade (real-time, certifiable)	✓	×	weak	varies	✓	✓
Neutral governance	✓	weak (Google)	weak (HF)	by definition no	× (NVIDIA)	✓
Standalone runtime value (not just data/training)	✓	×	✓	✓	✓	✓

Each existing effort satisfies a non-trivial subset of the requirements. None satisfies the full set. The conjunction of *semantically typed*, *cross-vendor on both sides*, *production grade*, and *neutrally governed* defines an empty point in the existing landscape. That point is the position RFL is designed to occupy.

A subtler observation, visible only when the five efforts are considered together, is that the field's coordination has been *over-attempted at the wrong layers* rather than *under-attempted in the aggregate*. There is no shortage of will to coordinate; there is a shortage of will to coordinate at the specific layer where coordination would have the largest network-effect leverage. ROS 2 coordinated transport; Open X-Embodiment coordinated training data; NVIDIA coordinated tooling; LeRobot coordinated research workflow. Each of these coordinations is real and useful. None of them is a coordination of *semantically meaningful action* between *competing model providers* and *competing embodiment providers*. That coordination is what RFL proposes.

3.7 2.7 The Implication for Design

Two design implications follow directly from the diagnostic above and govern the choices in § 3 and § 4.

First, **RFL must be additive to existing infrastructure rather than replacement**. ROS 2 is the right transport; Isaac Sim is a fine simulator; Open X-Embodiment data is excellent training material; LeRobot's developer experience is worth borrowing where compatible. The argument of this paper is not that the existing stack is wrong, but that it is incomplete, and that completing it requires a single new piece at a specific layer. The RFL specification is designed to compose with each of the above rather than compete with them.

Second, **the governance question is as load-bearing as the technical question**. The reason every existing effort fails to occupy the coordination position is, in three of five cases, governance rather than technology. A technically perfect specification housed under the wrong steward will not be adopted; a technically average specification housed under the right steward will be. § 7 of this paper develops the foundation governance structure in detail, but the design pressure is established here: the steward must be a neutral foundation, the spec must be permissively licensed in perpetuity, and the founding members must be drawn from genuinely competitive corners of both sides of the market.

The remainder of the paper takes these design pressures as constraints and develops the specification, network-effect architecture, and governance structure that satisfy them.

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4 § 3. RFL: First Principles

A specification proposed without first principles is a specification that drifts under commercial pressure. Every layer that has achieved durable cross-vendor adoption — POSIX, the C ABI, the ARM ISA, USB, HTTP — was anchored by a small set of design commitments that were stated before the technical details and that constrained the technical details. RFL adopts five such commitments. They are derived from the constraints established in §§ 1–2 (the combinatorial crisis on one side; the structural failures of adjacent efforts on the other), and every choice in § 4's specification can be traced back to them.

This section states the five principles, draws the structural isomorphism with the ARM ISA on which the RFL design pattern is modeled, and identifies the dimensions on which RFL must deliberately diverge from ARM because the physical-AI substrate differs from silicon in ways that compatibility intuition does not capture.

4.1 3.1 The Methodological Commitment

Before stating the principles, a methodological commitment: principles, not features. A design principle is a constraint that survives commercial pressure because adherents to the standard refuse to violate it even when individual short-term gains would be available from doing so. A feature is a capability that exists because someone wanted it. Principles can be evaluated against future situations the original drafters did not anticipate; features cannot. The history of every coordination-layer success in computing shows the same pattern: a small set of principles, vigorously defended, outperforms a large set of features, opportunistically accumulated.

The five principles below are the smallest set that, in our analysis, jointly forecloses each known failure mode of past abstraction-layer attempts. We have considered and rejected larger principle sets (the early POSIX standardization process accumulated more than a dozen guiding tenets and slowed correspondingly) and smaller principle sets (a three-principle articulation we initially favored failed to constrain the governance dimension and would have permitted vendor capture). Five is, on our analysis, the parsimonious count.

4.2 3.2 The Five Principles

4.2.1 Principle 1: Embodiment-Agnostic

The specification must avoid encoding the design choices of any particular embodiment. Action semantics must be expressible across tendon-driven, pneumatic, hydraulic, and as-yet-unknown actuation modalities; across four-finger, five-finger, six-finger, and parallel-jaw end effectors; across anthropomorphic and non-anthropomorphic kinematic structures.

This is the foundational constraint and the easiest one to violate. Almost every existing robotics convention is, on examination, quietly encoded with the assumptions of a dominant embodiment class. The Feix grasp taxonomy assumes the human hand; ROS 2's `geometry_msgs` family assumes Cartesian-end-effector control; the dominant teleoperation interfaces assume two-handed bimanual operation. RFL must not bake in any such assumption.

The operational test: a draft of the specification must be reviewable by someone whose only experience of robotics is a non-anthropomorphic platform — say, a pneumatic six-finger hand or a soft-bodied actuator — and they must find the semantics expressible in their world without translation through anthropomorphic intuition. If they cannot, the spec has smuggled in an embodiment assumption and must be rewritten.

4.2.2 Principle 2: Compositional

Skills are atomic primitives that compose, under a defined algebra, into higher-level behaviors. The specification defines the composition rules, not merely the primitive set.

A skill ISA that is only a list of primitives is a skill *library*, not a skill *language*. The difference matters because the value of an abstraction layer grows multiplicatively, not linearly, with the expressiveness of its composition rules. ARM's enduring success rests on its instruction set being composable into procedures, libraries, and operating systems through a small set of calling conventions and addressing modes; the primitives matter less than the composition rules built atop them.

RFL's Skill ISA defines, in addition to the primitive operations, the rules by which sequential, conditional, and concurrent composition is expressed, the way pre- and post-conditions of composed skills are derived from their components, and the algebra by which the safety envelope of a composition is inherited from its components. § 4.4's worked example exercises these composition rules in a single multi-step manipulation; § 4.1 enumerates the primitive set.

4.2.3 Principle 3: Verifiable

Every skill carries pre-conditions, post-conditions, and a safety envelope expressible in a machine-checkable form. Composition rules preserve verifiability.

The verification principle distinguishes RFL from an API. An API dispatches function calls; a verifiable abstraction layer additionally promises that, if the pre-conditions hold and the skill executes to its post-conditions, certain safety-relevant invariants will hold throughout the execution. This promise is what makes the Certification reinforcement loop (§ 5.4 L4) and the Insurance reinforcement loop (§ 5.4 L8) possible. Without verifiability, RFL cannot interoperate with the regulatory and actuarial institutions that physical-world deployment ultimately requires.

The verification principle imposes design discipline on the primitive set. A primitive whose safety envelope cannot be expressed compactly is rejected, regardless of how useful the primitive would otherwise be. This is a real cost; some genuinely useful operations (notably those involving deformable-object manipulation) will lie at the edge of expressibility and require iterative spec refinement. We accept the cost because the alternative — admitting unverifiable primitives — corrupts the certifiability of every composition that includes them.

4.2.4 Principle 4: Provider-Neutral

No vendor's preferred way of doing things has special status in the specification. The specification's evolution is governed by procedures that treat Founding Members and late entrants identically once admitted.

Provider neutrality is governance expressed at the level of the spec. It is the technical instantiation of the foundation governance principles developed in § 7. The principle has two operational tests: first, an outside reviewer should be unable to identify which Founding Member proposed any given primitive by inspecting the spec; second, the spec's evolution procedure (voting, comment periods, breaking-change moratoria) should produce the same outcome regardless of which member proposes the change.

This principle is what foreclosed the otherwise tempting design choice of letting the most technically capable Founding Member contribute a head-start spec and have other members ratify it. The technical benefits of such an approach (faster spec maturity, fewer compromises) are real, but the perception of vendor capture — and the actual erosion of cross-vendor adoption that perception would produce — is fatal. The compromise we adopt instead is that the v0.1 reference specification is drafted by the RFL Foundation's neutral spec-editor staff, with proposals from any Founding Member receiving equal procedural treatment.

4.2.5 Principle 5: Forward-Compatible

The specification is designed to extend without breaking. Future VLA capabilities and future embodiment classes must be expressible in extensions; the v1.x line must never deprecate v1.0 functionality.

The forward-compatibility principle is the discipline that distinguishes a standard from a moving target. Backward compatibility within a major version is not optional. New primitives, new composition rules, and new metadata fields enter the spec through additive extensions, organized into a versioned extension registry analogous to RISC-V's modular extensions or x86's CPUID feature bits. A VLA that targets RFL v1.0 in 2027 must continue to work against RFL-compliant embodiments shipped in 2032; an embodiment certified against RFL v1.0 must continue to receive VLA support indefinitely.

This principle imposes a non-trivial intellectual cost on the v0.1 design: the drafters must anticipate which extensions are foreseeable and structure the core spec so that those extensions can be added without disturbing existing semantics. We have used the design heuristic of *capability discovery* (every RFL-compliant component exposes its capabilities through a structured manifest) and *graceful degradation* (a VLA that requires a capability the embodiment lacks must receive a clear failure mode, not undefined behavior). § 4.3 develops both mechanisms in detail.

4.3 3.3 Isomorphism with the ARM ISA

The five principles, taken together, place RFL in structural correspondence with the ARM Instruction Set Architecture. The correspondence is not metaphorical; it is point-for-point.

RFL element	ARM ISA element
Skill ISA primitives	Instruction set primitives
Skill composition rules	Calling conventions, ABI
Pre/post-condition contracts	Architectural invariants (e.g., register state preservation)
Driver Interface	Memory model, peripheral I/O conventions

RFL element	ARM ISA element
Capability discovery manifest	CPUID-equivalent feature reporting
Extension registry	ARM architecture profiles (A/R/M, plus extensions)
Foundation governance	ARM Holdings as steward
Compliance test suite	ARM architecture validation suite

The correspondence is generative, not accidental. The first principles in § 3.2 were derived in part by asking, for each ARM ISA design choice that proved durable: *what was the structural reason that choice succeeded, and what is the physical-AI analog of that reason?* Embodiment-agnosticism corresponds to ARM's silicon-agnosticism (the ISA does not specify how the chip is fabricated); compositionality corresponds to ARM's procedure-and-library compositionality (the ISA defines how procedures call each other); provider-neutrality corresponds to ARM Holdings' decision not to manufacture chips itself; forward-compatibility corresponds to ARM's discipline of architectural-version stability across decades.

The historical evidence is strong. ARM's enduring position in silicon emerged precisely from disciplined adherence to a small set of principles that resisted commercial pressure to deviate. Two specific moments are illustrative: ARM's refusal to acquire chip-fabrication capacity in the early 2000s (which would have grown short-term revenue but destroyed the neutrality that anchored vendor adoption); and ARM's decision in the late 2010s to extend the architecture (ARMv8, ARMv9) through additive features rather than breaking changes (which preserved the multi-decade application binary base). RFL's principles 4 and 5 are direct descendants of those decisions.

4.4 3.4 Where RFL Must Diverge from ARM

The isomorphism, however, is not complete. The physical-AI substrate differs from silicon along four dimensions that the ARM design did not need to address and that RFL cannot inherit unchanged.

Safety as architectural concern. ARM's instructions do not physically harm. A buggy ARM program can hang the machine, corrupt data, or leak information, but it cannot crush a human hand. RFL's primitives can. The verifiability principle (Principle 3) and the safety envelope mechanism it requires have no ARM analog and must be designed from first principles for physical action. § 4.2's translation layer and § 4.4's worked example show the safety envelope's structure.

Two-sided market urgency. ARM emerged into a market where chip vendors and OS providers were not racing each other to vertical integration. The silicon ecosystem had time to consolidate around the ISA. RFL faces a market in which vertical integration (Figure–Helix, Tesla–Optimus, NVIDIA–GR00T) is already underway and accelerating; the window for a horizontal coordination layer is narrower than ARM's was, and the founding-member commitments must be secured before vertical lock-in ossifies. This urgency shapes the cold-start strategy in § 5.3 and the consortium recruitment plan in § 9.

Data gravity as a moat input. Silicon does not generate data that improves silicon. Physical AI does: every executed action becomes training data for the next generation of translation models. RFL's design must therefore include the data-flow architecture that captures and shares this learning across the ecosystem (the L1 Data reinforcement loop in § 5.4) without violating the privacy or commercial interests of the participating Founding Members. § 4.5's conformance test suite and § 7's data-sharing governance both descend from this concern.

Multi-jurisdictional regulatory exposure. ARM faced one significant regulatory question (export controls on advanced cores). RFL faces a dense thicket: ISO 10218 and 13482 on industrial-robot safety, the emerging EU AI Act provisions on autonomous systems, sector-specific medical-device regulation for healthcare embodiments, and a patchwork of national rules on AI accountability. The forward-compatibility principle (Principle 5) acquires a regulatory dimension that ARM never had: the spec must be extensible not only to absorb future capabilities but to absorb future regulatory frameworks without architectural disruption.

Each of these divergences makes RFL harder to design than ARM was. None of them invalidates the structural correspondence. The five principles remain the spine; the divergences are constraints on how the spine is fleshed out.

4.5 3.5 How the Principles Constrain the Specification

The principles are not aspirational; they are operational. § 4's specification is constructed under their constraint in the following specific ways.

The Skill ISA primitive set is restricted to operations whose pre/post-conditions and safety envelopes are expressible in the verification language (Principle 3) and whose semantics are stable across at least three structurally different embodiments in the v0.1 reference implementations (Principle 1). The Translation Layer's canonical action space is parameterized along axes that admit any kinematic and actuation modality currently known to be commercially feasible, rather than along axes that are convenient for a specific class of robots (Principle 1). The Driver Interface contract is designed so that any robot whose firmware exposes the contract can host any RFL-compliant VLA, without negotiation specific to that VLA (Principle 4). Every primitive carries a version stamp from its first appearance, and the v1.x evolution policy explicitly forbids retracting a stamped primitive (Principle 5). The composition rules are defined such that the safety envelope of a composed skill is computable from the safety envelopes of its components, enabling certification at composition granularity rather than requiring per-composition reverification (Principles 2 and 3).

These constraints make § 4 longer and more conservative than a feature-driven specification would be. They are the deliberate cost of building a layer that can survive the commercial pressure it will encounter, in the same way that ARM's disciplined architectural conservatism imposed costs in 1985 that paid off across the four decades that followed.

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5 § 4. The RFL Specification v0.1

This section presents the Robot Foundation Layer specification at version 0.1. It is deliberately narrow in scope: it specifies the abstraction layer between Vision-Language-Action (VLA) intelligence and robotic embodiment, and nothing else. Layers above the spec (task planning, world modeling) and below it (real-time motor control, sensor drivers) are explicitly out of scope, in accordance with the Layer Discipline principle (§ 5.8, derived from Principle 4). The spec defines three layers: a Skill Layer providing semantically typed action primitives and their composition algebra; a Translation Layer providing a canonical, embodiment-agnostic action representation; and a Driver Interface defining the contract between the Translation Layer and embodiment-specific firmware.

The spec is presented at a level of detail sufficient to fix the architectural commitments and to enable competent implementers to begin work. Full implementation detail — every type signature, every JSON schema, every error code — is deferred to the companion GitHub repository, where the spec lives as a versioned document subject to the evolution procedure described in § 7.

5.1 4.1 The Skill Layer

The Skill Layer defines fifty base primitives organized into seven categories, plus the compositional algebra that combines them into multi-step manipulations. The primitive count is itself a design commitment: too few primitives forces VLAs to express common manipulations through awkward compositions, while too many primitives encourages embodiment-specific encoding (Principle 1). Fifty is the count at which, in our v0.1 design review with the prospective Founding Consortium, every commonly cited industrial manipulation could be expressed in three or fewer composition steps, with no primitive applying to only a single embodiment class.

The categories and representative primitives:

Category	Count	Representative primitives
Grasping	10	`grasp_power`, `grasp_precision`, `grasp_pinch`, `grasp_lateral`, `grasp_hook`, `grasp_cylindrical`, `release`, `regrasp`, `transfer_grasp`, `adjust_grip`
Translation & placement	8	`place`, `push`, `pull`, `slide`, `drag`, `stack`, `retract`, `advance`
Rotation & orientation	6	`rotate`, `twist`, `screw_in`, `screw_out`, `reorient`, `align`
Insertion & connection	8	`insert`, `seat`, `mate`, `snap`, `latch`, `clip`, `lock`, `plug`
Continuous manipulation	8	`pour`, `wipe`, `fold`, `unfold`, `smooth`, `crease`, `drape`, `wrap`

Category	Count	Representative primitives
Inspection & sensing	6	`probe`, `palpate`, `scan`, `measure_force`, `measure_compliance`, `verify_state`
Multi-effector coordination	4	`bimanual_handoff`, `bimanual_hold_and_act`, `multi_finger_regrasp`, `cooperative_carry`

Each primitive is specified through five fields: a *type signature* (typed inputs and outputs), *pre-conditions* (environmental and state predicates that must hold before invocation), *post-conditions* (predicates that hold upon successful completion), a *safety envelope* (force bounds, motion bounds, and abort triggers that hold throughout execution), and a *capability requirement* (the embodiment capabilities, declared through the Driver Interface, that the primitive requires for execution).

As a concrete example, the `grasp_pinch` primitive is specified as follows:

```
primitive grasp_pinch:
  inputs:
    target:       Object3D           # geometry + pose + material descriptor
    force_budget: Force[N]           # default 5.0 N, max 50.0 N
    grasp_pose:   Pose6D | "auto"    # default "auto" (planner-derived)
  outputs:
    result:       {success, slip, force_exceeded, unreachable}
    final_pose:   Pose6D             # actual grasp pose achieved
    telemetry:    GraspTelemetry     # tactile + force trace
  preconditions:
    - target.is_visible_to(embodiment.perception)
    - target.estimated_mass <= embodiment.capabilities.payload_grasp_pinch
    - embodiment.state.end_effector_free
  postconditions:
    - if result == success: object_held(target, mode=pinch)
    - if result == slip:    embodiment.state.end_effector_free
  safety_envelope:
    - peak_force <= force_budget * 1.2         # 20% transient margin
    - peak_velocity <= embodiment.limits.v_grasp
    - abort_if: target.crush_indicator > 0.8
  capability_requirement:
    - {pinch_grasp: required}
    - {tactile_sensing: preferred, level: "binary_contact" or higher}
```

The specification structure is uniform across all fifty primitives, which is what enables the verifiability principle (§ 3.2 Principle 3): a tool can mechanically check that every primitive in a composition has its pre-conditions established by predecessors and that the composition's safety envelope is the proper composition of the individual envelopes.

Compositional algebra. The Skill Layer defines five composition operators, sufficient to express the manipulation workflows we examined across the v0.1 design review.

Operator	Syntax	Semantics
Sequential	`A ; B`	Execute `A` to success, then execute `B`; if `A` fails, composition fails
Conditional	`A ? B : C`	Execute predicate `A`; on true execute `B`, on false execute `C`
Concurrent	`A` ‖ `B`	Execute `A` and `B` simultaneously; safety envelope is union; composition succeeds when both succeed
Reactive	`A until trigger`	Execute `A` until `trigger` fires, then halt `A` cleanly
Bounded retry	`try A retries n`	Execute `A`; on failure retry up to `n` times before failing the composition

Composition rules preserve verifiability. The pre-condition of a composition is the pre-condition of its first-executed component; the post-condition is the post-condition of the last component along the executed path; the safety envelope is the union of the components' envelopes for `;` and `?`, and the strict intersection for ‖ (any envelope violation by either concurrent action aborts both). Bounded retry inherits the envelope of `A` unchanged but admits multiple attempts within it.

A subtlety worth noting: the Concurrent operator's strict-intersection safety semantics is deliberately conservative, because two concurrent actions whose envelopes individually permit certain forces may, when executed together, produce force interactions that neither envelope anticipated. The conservative choice forecloses a class of compositions that would, in some embodiments, be safe; we accept the cost as the price of mechanical verifiability.

5.2 4.2 The Translation Layer

The Translation Layer converts Skill Layer invocations into a canonical, embodiment-agnostic action representation that the Driver Interface can route to embodiment-specific firmware. It is the spec's most architecturally novel layer, occupying the position that ROS 2 does not (semantic translation, not just message transport) and that single-model VLAs cannot (cross-embodiment routing, not just internal model dispatch).

The canonical action representation is a tuple:

```
CanonicalAction = (
  pose_trajectory:     SE(3) sequence with timestamps,
  force_trajectory:    Force[6] sequence aligned with pose_trajectory,
  tactile_target:      TactileManifold | None,
  proprioceptive:      JointConfig | None,
  capability_flags:    set of CapabilityFlag,
  confidence:          DistributionOverSuccess,
  safety_envelope:     SafetyEnvelope (inherited from Skill Layer)
)
```

Five aspects of this representation are load-bearing.

First, the pose and force trajectories are expressed in the *task frame*, not in any embodiment-specific frame. The Driver Interface's calibration handshake (§ 4.3) is responsible for mapping task-frame coordinates into the embodiment's native frame at execution time. This indirection is what allows the same canonical action to drive radically different embodiments without retargeting at the VLA layer.

Second, the tactile target is expressed in the TactileManifold abstraction — a topological description of contact patterns rather than a specific tactile-sensor output format. An RFL-compliant embodiment whose tactile sensors produce four-channel binary contact data can describe its observations in TactileManifold terms; an embodiment with high-resolution GelSight-class visuotactile output can describe its richer observations in the same terms with higher resolution. The abstraction is leaky in a controlled way: VLAs can request a minimum manifold resolution through capability flags, and the Translation Layer's capability negotiation will route the action to embodiments that meet the requirement (Principle 5: graceful degradation).

Third, the proprioceptive field is *optional*. Many Skill Layer primitives — particularly grasping and continuous manipulation — do not require joint-level specification, because the Translation Layer can derive a kinematically feasible joint trajectory from the task-frame trajectory and the embodiment descriptor. The field is present for the cases where the VLA has a specific reason to control joint configuration (avoiding obstacles, matching a learned policy's training distribution).

Fourth, the confidence field carries a distribution over outcomes rather than a point estimate. This is the input the Driver Interface needs to set appropriate safety margins: a high-confidence action can execute with aggressive timing, a low-confidence action triggers conservative timing and richer telemetry capture.

Fifth, the safety envelope is inherited unchanged from the Skill Layer composition. The Translation Layer is responsible for ensuring that the canonical action it produces will not, when executed, violate the envelope; if no kinematically feasible trajectory exists within the envelope, the Translation Layer returns a failure to the Skill Layer rather than relaxing the envelope.

Retargeting algebra. The mapping from a Skill Layer invocation to a canonical action depends on the target embodiment. The Translation Layer is parameterized by an *embodiment descriptor* (loaded from the Driver Interface at session initialization) and applies a deterministic retargeting algorithm:

```
canonical_action = retarget(
  skill_invocation,
  embodiment_descriptor,
  current_environment_state
)
```

The retargeting algorithm is itself versioned and open-source, with the v0.1 reference implementation written in Rust and exposed through both a Python binding (for ML integration) and a C binding (for embedded integration). The algorithm's behavior is fully specified in the spec document, so that any compliant implementation produces, for the same inputs, identical canonical actions. This determinism is what makes the conformance test suite (§ 4.5) possible.

This determinism carries a deliberate boundary worth stating plainly. The force-related envelope bounds, most consequentially the minimum holding force that keeps a grasped object from slipping, are derived from a declared friction coefficient and a schematic capacity model, and are never measured at runtime: a runtime measurement would make the retargeting output depend on sensor noise and break the byte-level reproducibility the conformance regime rests on. Real slip depends on contact-patch deformation, surface contamination, and load history that a single declared coefficient does not capture. RFL accepts this as the price of determinism: the contract fixes a reproducible, auditable lower bound, and an embodiment that needs a tighter guarantee layers runtime force sensing above the contract rather than inside it.

5.3 4.3 The Driver Interface

The Driver Interface defines the contract between the Translation Layer and embodiment-specific firmware. It has three components: extended kinematic description, capability discovery, and a runtime action protocol.

Extended kinematic description. RFL extends the URDF/MJCF kinematic-description standard with additional fields required for the spec's purposes:

- Tactile sensor placement and modality (location on the kinematic tree, manifold resolution, native data format)
- Actuation modality (electric, pneumatic, hydraulic, tendon-driven) and its associated dynamic properties (bandwidth, hysteresis, backlash)
- Proprioceptive sensor characterization (joint encoder resolution, force-torque sensor placement and bandwidth)
- Safety-relevant kinematic limits (joint velocity ceilings, end-effector workspace bounds, collision-detection model)
- Calibration metadata (when the embodiment was last calibrated, against which reference, with what residual error)

The extension is additive — a standard URDF parser produces a usable but incomplete embodiment descriptor; an RFL-aware parser produces the full descriptor. This composability with existing tooling is itself a Principle 4 commitment.

Capability discovery. Every RFL-compliant embodiment exposes, at session initialization, a manifest declaring which Skill Layer primitives it can execute, with what fidelity, and under what conditions. The manifest is structured:

```
capability_manifest:
  primitives_supported:
    grasp_pinch:
     fidelity:      "production"        # "experimental" | "stable" | "production"
      success_rate:    0.95                 # empirical on conformance test
      latency_p95:     0.8 seconds
      payload_max:     0.5 kg
      object_size_min: 5 mm
      object_size_max: 80 mm
      tactile_level:   "binary_contact"
    grasp_power:
      fidelity:        "production"
      ...
  capabilities_declared:
    {pinch_grasp, power_grasp, lateral_pinch, tactile_sensing,
     force_sensing_3axis, dual_arm: false}
  extensions:
    [rfl-tactile-highres-v0.1]               # opt-in extensions
```

The manifest is the basis for capability negotiation between the VLA and the embodiment. A VLA targeting `grasp_pinch` against an embodiment whose manifest declares `fidelity: experimental` knows to set conservative confidence; an embodiment whose manifest does not declare `pinch_grasp` capability is matched against a different primitive (`grasp_lateral` is the canonical fallback) through the Translation Layer's retargeting algorithm.

Runtime action protocol. At runtime, canonical actions are dispatched to the embodiment through a ROS 2 action server interface. This is a deliberate compatibility commitment with the dominant transport standard (§ 2.1): every RFL-compliant embodiment is, by virtue of compliance, exposed as a set of ROS 2 action servers, one per supported primitive. The action server's goal message carries the canonical action; the feedback messages carry execution telemetry at a declared cadence; the result message carries the outcome plus a structured execution trace. The trace structure is itself standardized — it is consumed by the conformance test suite and by certification tooling, and its standardization is what enables third-party audit of RFL-compliant behavior.

The choice of ROS 2 as the wire protocol does not preclude alternative transports (gRPC, MQTT, DDS-other) for embodiments that do not run ROS 2 internally; the spec defines a wire-format-neutral message structure and provides reference bindings for each major transport. ROS 2 is the default because it is the deployed standard, not because it is privileged in the spec.

5.4 4.4 Worked Example: Cable-and-Connector Insertion

We illustrate the layers together with a worked example drawn from a class of industrial manipulation tasks that motivated the early phase of this work: picking up a flexible cable terminated by a connector and inserting it into a receptacle. The task is structurally non-trivial — it combines compliant-object grasping (the cable is flexible), force-controlled insertion (over-insertion damages pins), and visual servoing (the receptacle pose has tolerance under millimeter scale).

A VLA targeting this task emits a Skill Layer composition:

```
task: insert_cable_into_connector
composition:
  grasp_pinch(target=cable_end_visual, force_budget=3.0N) ;
  reorient(target=connector_alignment_pose) ;
  align(reference=connector_axis, tolerance=2mm) ;
  insert(target=connector_socket, force_budget=8.0N,
         abort_if=force_exceeded) until seated_indicator(force>=4N, depth>=8mm)
```

The Translation Layer expands this composition into a sequence of canonical actions, each parameterized for the target embodiment. For a four-finger tendon-driven hand (e.g., Allegro), `grasp_pinch` resolves to a thumb-and-index opposing grasp with joint trajectory derived from the task-frame target pose; the safety envelope's 3N force budget is expressed as bounded joint torques. For a four-finger direct-drive hand (e.g., LEAP), the same skill resolves to the same opposing geometry but with different torque expressions reflecting the embodiment's actuation properties. For a six-finger pneumatic hand (e.g., Bridgestone TETOTE), the skill resolves to a front-two-finger pinch with the force budget expressed as a pneumatic pressure setpoint (approximately 0.3 MPa for a 3N target force, calibrated through the embodiment descriptor).

Each resolved canonical action is dispatched to the embodiment's ROS 2 action server. The Driver Interface translates the canonical action into motor commands appropriate to the embodiment's actuation. Tactile feedback flows back through the Translation Layer, where it is converted into the TactileManifold representation and made available to the VLA for closed-loop control.

The crucial observation: the VLA's composition is identical across all three embodiments. The VLA does not need to know whether its target is tendon-driven, direct-drive, or pneumatic; it does not need to know whether the hand has four fingers or six. The translation is performed by the layer, not by the model. This is the property that, multiplied across the realistic deployment matrix described in § 1.1, collapses the $N \times M$ integration cost to additive.

A note on where this composition originates, since it carries the standard's central dependency assumption. The Skill Layer composition above is emitted by the planner tier: the hierarchical-VLA, Code-as-Policies, or "System-2" reasoning layer that interprets a task and produces structured sub-goals. It is not emitted by an action-token policy that outputs low-level continuous motor commands at tens of hertz. Those action-token VLAs ($\pi 0$, OpenVLA, GR00T) sit below the canonical-action boundary, close to the Driver Interface, and RFL does not ask them to produce symbolic compositions. The natural emitter of the Skill ISA is the general planner archetype that a frontier model already provides. That this is realizable rather than aspirational is established by the project's demand-side existence proof, in which a current frontier model, given the Skill ISA schema but not the embodiment, emits valid full-spec Skill ISA on every task in a held-out set of novel manipulation problems. RFL therefore rests on the planner tier producing structured intent, which is the function of that tier, and not on action-token models acquiring a new output modality.

A second observation: the composition's `insert` step uses the Reactive operator (`until seated_indicator(...)`), which allows the action to terminate cleanly when tactile feedback indicates seating. The Skill Layer's compositional algebra makes this reactive behavior expressible without requiring the VLA to micromanage timing — the embodiment-side implementation of `seated_indicator` is part of the Driver Interface, and its specific implementation varies by embodiment (force-threshold-based for one, deformation-pattern-based for another) but its semantics are uniform.

The example is one of many that the v0.1 design review exercised; the GitHub reference implementation will ship with a curated example library covering each of the seven primitive categories and each of the five composition operators.

5.5 4.5 Conformance and Certification

A specification without a conformance regime is a recommendation, not a standard. RFL ships v0.1 with an open-source conformance test suite that defines, mechanically, what it means for an implementation — VLA or embodiment — to be RFL-compliant.

The suite has four test classes.

Capability conformance. For every primitive an embodiment declares in its capability manifest, the suite executes a battery of test invocations against a calibrated test bench and measures success rate, latency, and safety-envelope adherence. The empirical results become the values populating the manifest; an embodiment's claim of `success_rate: 0.95` is grounded in measurement, not assertion. The test bench specifications are themselves public, so that independent labs can reproduce conformance results.

Composition conformance. The suite executes a fixed library of multi-primitive compositions (sequential, conditional, concurrent, reactive, retry) and verifies that the embodiment's behavior remains within the declared safety envelope of the composition and that the composition's outcome matches the declared semantics. A composition that succeeds on the individual primitives but produces a safety-envelope violation in composition fails the conformance test, regardless of the individual primitive results.

Capability discovery conformance. The suite queries the capability manifest, then attempts primitives that the manifest does and does not declare; the embodiment must succeed on declared primitives at the declared rate and must cleanly reject (not silently fail, not unsafely attempt) primitives it does not declare.

Safety conformance. The suite deliberately invokes primitives at the edge of and beyond their declared safety envelopes (in simulation; certain physical tests are run only on instrumented test articles). The embodiment must abort within the declared response time and must leave the system in a safe state after the abort.

An implementation passing all four classes at a declared spec version is *RFL-compliant* at that version and may use the RFL™ trademark on packaging and documentation per the trademark policy described in § 7. Implementations passing partial classes are *RFL-conformant-subset* and may declare which classes they pass, but may not use the trademark.

The conformance regime is what makes the Certification reinforcement loop (§ 5.4 L4) and the Insurance reinforcement loop (§ 5.4 L8) operational. A notified body certifying a deployment against ISO 10218 or ISO 13482 has a defined evaluation surface — the RFL conformance results plus the embodiment's deployment context — rather than having to evaluate the deployment from first principles. An insurer pricing liability against an RFL-compliant deployment has a defined risk profile — the empirical conformance results across the deployed primitives — rather than relying on per-deployment underwriting.

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6 § 5. Network Effects and Ecosystem Design

A standard architecture proposal stops at the boundary of the artifact: define the spec, ship a reference implementation, hope adoption follows. This is insufficient for the Robot Foundation Layer (RFL). Because RFL sits between two rapidly evolving sides — Vision-Language-Action (VLA) foundation models on one side, robotic embodiments on the other — its long-run viability depends not on technical elegance alone but on whether it can become the **substrate around which both sides organize their mutual investment**. This section formalizes the network-effect structure of RFL, derives its minimal viable configuration, identifies the nine reinforcement loops that compound into a long-term moat, and articulates the governance principles required to keep the layer trustworthy as it scales.

6.1 5.1 The Two-Sided Market Structure of Physical AI

The physical-AI ecosystem of 2026 is, in the strict sense of Rochet and Tirole (2003), a two-sided market. On the N side sit VLA providers — Physical Intelligence ($\pi 0$, $\pi 0.5$), OpenVLA, Skild AI, Google DeepMind Gemini Robotics, NVIDIA GR00T N1, Figure Helix, and the rising tide of academic checkpoints. On the M side sit embodiment providers — Figure, Tesla, 1X, Unitree, Apptronik, Bridgestone (TETOTE), K-scale Labs, and the open-source ecosystem around LEAP Hand, Allegro Hand, and Inspire Robotics. Each side derives indirect utility from the variety, quality, and trustworthiness of the other: a VLA without a body cannot act, and an embodiment without intelligence cannot decide.

What makes the present moment pathological is that the binding between the two sides is bilateral and bespoke. Each (VLA, embodiment) pair requires a custom integration that today consumes an estimated three-to-six engineer-months and a per-pair cost in the low to high hundreds of thousands of dollars (see § 1). With approximately nine credible VLAs and roughly thirty distinguishable embodiments now in commercial or near-commercial use, the field faces $O(N \times M)$ integration cost — a quadratic that suppresses experimentation, locks the market into vertically integrated stacks (Figure–Helix, Tesla–Optimus), and prevents the cross-pollination that has historically driven platform progress in computing.

RFL is proposed as the canonical translation layer that collapses this quadratic to additive cost. Once a VLA targets the RFL Skill ISA and an embodiment exposes the RFL Driver Interface, integration becomes the composition of two compliant endpoints rather than a bespoke engineering project. The resulting market structure is the classical two-sided platform: a neutral central spec coordinates multi-homing on both sides while declining to compete with either.

The critical design constraint, which we revisit throughout this section, is that **RFL must not vertically integrate**. The moment the Foundation behind RFL ships a VLA or a robot, it forfeits the neutrality that anchors both-sided trust. ARM's exit from Acorn was not incidental — it was the structural precondition for becoming the default architecture.

6.2 5.2 Bilateral Intelligence Between VLAs and Embodiments

A naive reading of RFL would treat it as an API: VLA pushes joint targets, embodiment executes. This reading misses the structural depth of the problem and the moat that depth enables. We borrow from the Bilateral Intelligence Spectrum first formalized in the context of personal-AI social networks (Hara, 2026) and adapt it to the VLA $\times$ embodiment substrate. The spectrum has five levels.

Level	Name	What it requires	Representative state in 2026
L0	No translation	Each pair runs in its own format; no interop	Figure–Helix $\rightarrow$ Figure-only hand
L1	Unilateral (VLA-aware)	VLA models multiple embodiments; embodiments are blind to the VLA	π0 trained across several arms
L2	Unilateral (embodiment-aware)	Embodiments expose a generic interface; VLA is blind to the body	ROS 2 action servers

Level	Name	What it requires	Representative state in 2026
L3	Inferred Bilateral	A translation layer holds explicit models of both sides and mediates between them	**RFL v1.0 target**
L4	True Bilateral	VLA and embodiment co-adapt at runtime, exchanging structural updates	**RFL v2.0+ horizon**

The strategic significance of L3 is that no shipped competitor occupies it. LeRobot and ROS 2 are L2 at best — they standardize the embodiment side but expect the model side to adapt downward. $\pi 0$ is the most successful L1 effort, internalizing many embodiments inside a single model but offering no externally usable abstraction. The L3 position is empty, and L3 is precisely the level at which the abstraction layer becomes architecturally indispensable rather than merely convenient.

The mathematics behind this asymmetry are not folk wisdom. Cremer and McLean (1985, 1988) proved, in the mechanism-design setting, that a designer with bilateral information can implement every outcome a unilateral designer can implement, plus an additional set unattainable to either side alone. Translated into our context: the value of high-fidelity embodiment modeling rises in the fidelity of VLA modeling, and vice versa. Formally, letting V denote translation value and I_X denote information held by side X:

$$V(I_{\text{VLA}}^{\text{high}}, I_{\text{emb}}^{\text{high}}) - V(I_{\text{VLA}}^{\text{high}}, I_{\text{emb}}^{\text{low}}) > V(I_{\text{VLA}}^{\text{low}}, I_{\text{emb}}^{\text{high}}) - V(I_{\text{VLA}}^{\text{low}}, I_{\text{emb}}^{\text{low}})$$

This is the supermodularity condition. It implies that the marginal return on RFL's investment in either side's modeling is strictly increasing in the quality of the other side's modeling. Empirically, this is consistent with the multi-embodiment ablations reported by Physical Intelligence (2024) and the cross-embodiment transfer results of Open X-Embodiment (Google et al., 2023), both of which show super-linear scaling rather than additive transfer. We defer the formal derivation to Appendix C.

6.3 5.3 The Atomic Network: Minimal Viable Configuration

The Cold Start literature establishes that two-sided platforms succeed by first proving value within an "atomic network" — the smallest configuration that exhibits the platform's defining property (Chen, 2021). For RFL the defining property is *translation*, and the atomic network must therefore demonstrate that a single high-level instruction, expressed once, executes correctly across multiple non-trivially different embodiments under multiple non-trivially different intelligences. The minimum is **two VLAs** $\times$ **two embodiments** $\times$ **one environment** $\times$ **one task**:

```
VLA         : π0 (Physical Intelligence) + OpenVLA (Stanford OSS)
```

```
Embodiment   : LEAP Hand (CMU OSS, ~$2k) + Allegro Hand (Wonik commercial, ~$20k)
Environment  : standard table, calibrated lighting, Aruco markers
Task         : pick-and-place (4 objects × 3 placement targets)
Pass         : >80% success across all four (VLA, embodiment) combinations
Budget       : ~$50k, six months, two engineers
```

Three properties of this configuration matter. First, all four pairs must be **cross-vendor**: same-vendor combinations (e.g., $\pi 0$ with two CMU hands) do not demonstrate neutrality and therefore do not anchor consortium trust. Second, the choice of LEAP and Allegro — both four-finger but with different actuation, different control bandwidth, and different proprioceptive characteristics — exercises the translation layer without overwhelming the engineering budget. Third, the artifact of the PoC is not the code; it is the **video**. A one-minute demonstration in which the same instruction routes through RFL into two visibly distinct hands collapses the explanation cost of the entire RFL pitch to zero, and is therefore the gating deliverable for every downstream step (consortium recruitment, seed financing, CTO commitment).

A more ambitious atomic network — three VLAs × three hands, or four × four — is a strictly worse choice during this phase. It increases failure probability, dilutes the message, and consumes engineering hours that should instead be invested in spec maturation. The Outband Cold Start playbook (Hara, 2026, ch. 36) emphasizes this point: *minimal* atomic networks are not under-ambitious, they are the disciplined application of the cold-start theorem.

A two-sided cold start requires identification of the "hard side" — the side that, once committed, is slow to defect and thereby anchors the soft side. The intuitive answer for RFL is to anchor VLA labs first: they are technically central, intellectually adjacent to academia, and verbally enthusiastic about standards. The intuition is wrong. VLA labs are *easy* to recruit — the marginal cost of adding RFL support to an existing model is small, the experimentation is reversible, and labs benefit from any standard that broadens their model's reach. **Robot OEMs are the hard side.** Their commitment binds hardware design decisions made years in advance; their procurement cycles take quarters; and once they choose a software substrate they rarely re-platform. Five OEM commitments, locked in early, will pull twenty VLA integrations behind them. The reverse does not hold.

6.4 5.4 Nine Reinforcement Loops

A common error in network-effect design is to identify the single dominant loop and rely on it. ARM is often described, anachronistically, as having won on "ISA compatibility." In fact ARM's defensibility derived from at least five independently compounding loops — developer mindshare, silicon-vendor lock-in, mobile-first thermal advantage, ABI inertia, and standards-body neutrality — each of which was independently sustainable but which together became immovable. RFL is designed from the outset around nine such loops, sequenced in three temporal waves:

#	Loop	Mechanism	Activation gate	Wave
L1	Data	Every executed action improves the translation model	10+ connected embodiments	Immediate
L2	Skills	Standard-ISA skills accumulate as a public library	100+ certified skills	Immediate
L3	Developers	Learn once, deploy many $\rightarrow$ developer count compounds	Tutorials + community	Immediate
L4	Certification	Certified skills carry industrial trust; trust drives adoption	Notified body MoU	Mid-term
L5	Benchmarks	Common evaluation enables apples-to-apples comparison and press coverage	Benchmark suite v1	Mid-term
L6	Simulation	Shared sim environments amortize the sim2real cost	Isaac/MuJoCo bindings	Mid-term
L7	Marketplace	Skill royalties create supply-side economics	Payment infrastructure	Long-term
L8	Insurance	RFL-compliant systems carry lower liability premiums	Carrier partnerships	Long-term
L9	Regulatory	RFL compliance becomes the de-facto path to ISO/AI-Act adherence	Government engagement	Long-term

The temporal phasing is not optional. L1–L3 activate from day one of public release because they only require code and community. L4–L6 require external institutional partners and therefore cannot precede a credible spec. L7–L9 depend on commercial volume and regulatory maturity that themselves require years. The discipline implied by this matrix is that Phase 1 fundraising arguments must rest on L1–L3 (which are concretely achievable in 18 months), while Series B and beyond can credibly reference L4–L9 in a way that elevates valuation by an order of magnitude.

Crucially, the loops compound *multiplicatively*, not additively. Letting $L_i(t)$ denote the strength of loop i at time t and w_i the weight assigned to each loop:

$$M(t) = \prod_{i=1}^{9} L_i(t)^{w_i}, \quad \sum_{i=1}^{9} w_i = 1$$

The multiplicative form has two consequences that are often misunderstood. On the upside, acceleration in any loop magnifies the value of every other loop — a marketplace breakthrough (L7) raises the return on certification investment (L4) because certified skills now monetize. On the downside, complete failure of any single loop drives total moat strength toward zero. This is why ARM's near-collapse under Intel's mobile push in 2015–2017 was reversed not by doubling down on ISA compatibility but by reinforcing every adjacent loop in parallel.

6.5 5.5 RFL Master Equation

Synthesizing the above into a single formalization, the long-run value of RFL admits the following characterization:

$$\text{Value}(t) = \int_0^t \beta \cdot N_{\text{VLA}}(s) \cdot N_{\text{emb}}(s) \cdot |\Sigma(s)| \cdot Q_{\text{tr}}(s) \cdot T_{\text{cert}}(s)\, ds$$

subject to the constraint set

$$\begin{aligned}
\text{Neutrality}(s) &\geq 0.9 \quad \text{(no single vendor exceeds 30\% governance share)}\\
\text{SpecStability}(s) &\geq 0.95 \quad \text{(backward-compatibility ratio)}\\
\text{IntegrationCost}(s) &\leq \$50k \quad \text{(per (VLA, embodiment) pair)}\\
\text{Safety}(s) &= 1 \quad \text{(ISO 10218/13482 compliance)}\\
\text{GovTransparency}(s) &= 1 \quad \text{(Foundation meetings public)}
\end{aligned}$$

where N_{VLA} and N_{emb} count compliant VLAs and embodiments, $|\Sigma(s)|$ the size of the certified-skill library, $Q_{\text{tr}}(s)$ the empirical translation quality (measured on the common benchmark suite from L5), $T_{\text{cert}}(s)$ the cumulative trust accrued through L4 certification, and β a scale coefficient calibrated against historical platform growth curves.

The equation makes three claims explicit. First, value compounds multiplicatively across the five drivers in the integrand — none is substitutable for another. Second, every constraint is a *binding* one: a violation of neutrality or safety does not merely degrade the integral, it collapses the cumulative trust on which $T_{\text{cert}}(s)$ depends, propagating backward through the integral and destroying historical value. Third, the integral form captures path dependence: late entrants cannot recover the data and trust that compound from early activation. This is the formal expression of why the 2026–2028 window is decisive.

6.6 5.6 Comparative Defensibility Analysis

To position RFL against the platforms it most resembles, we apply Helmer's (2016) Seven Powers framework. The result, summarized in Table 5.A, is that RFL is structurally capable of accumulating a deeper moat than ARM ever did — not because RFL is technically superior, but because the physical-AI substrate exposes more loops to capture.

Power	ARM	CUDA	USB	ROS 2	LeRobot	RFL
Scale Economies	High	High	Low	Low	Low	Medium
Network Economies	Medium	Medium	High	Medium	Low	**Very High**
Counter-Positioning	High	Low	Medium	Medium	Low	Medium
Switching Costs	High	High	Low	Medium	Low	**Very High**
Branding	Medium	High	Low	Medium	Medium	High (neutrality)
Cornered Resource	None	None	None	None	Hugging Face mindshare	Founding Consortium contracts
Process Power	Medium	High	Low	High	High	High

ARM's celebrated moat ran on Scale Economies, Switching Costs, and Counter-Positioning. RFL adds Network Economies (the two-sided structure has no analog in ARM's chip-side market), elevates Switching Costs (data, skills, and certification each ratchet independently), and introduces a Cornered Resource (long-term contracts with Founding Members that competitors cannot replicate). Most importantly, RFL's loops L4–L9 — certification, insurance, regulatory — have no ARM equivalent because silicon never required the institutional trust that physical-world action demands. The conclusion is not that RFL will inevitably succeed, but that the structural ceiling of its defensibility exceeds ARM's. Whether RFL approaches that ceiling depends on execution.

6.7 5.7 Strategic Dynamics: Cold Start, Tipping Point, Anti-Patterns

The transition from atomic network (Day 90) to tipping point (Year 2) is the most fragile passage in the trajectory. We adopt the dual strategy of "come for the tool, stay for the network": the RFL Translation Layer must offer standalone value — measurable integration-cost reduction — to any single (VLA, embodiment) pair, even before the network is dense enough for indirect effects to matter. This standalone value is what carries early adopters across the chasm during which network density is insufficient to attract them on indirect-utility grounds alone.

Three failure modes warrant explicit pre-commitment to avoid:

- **The Symbian pattern**: governance designed for maximal neutrality, but so consensus-bound that vertically integrated competitors out-execute it. RFL's countermeasure is a two-tier governance structure — a Founding Member majority for routine decisions, paired with a public community veto right for breaking changes — that preserves speed without forfeiting legitimacy.
- **The Yik Yak pattern**: rapid early growth without invested community management collapses. RFL's countermeasure is a constitutional governance document published on day one, binding the Foundation to user-welfare primacy before commercial pressure can erode it.
- **The Path pattern**: over-constrained design (Path's Dunbar 150-friend cap) suppresses growth before the network effect can compound. RFL's countermeasure is to design the spec as *extensible by construction* — modeled on RISC-V's modular extension philosophy — so that vertical-specific skill sets can extend the standard without forking it.

Tipping is identifiable, ex ante, by three converging indicators: Founding Consortium reaches seven members, the GitHub reference implementation crosses 10k stars, and the foundational spec accumulates 50+ academic citations. The historical analog — ARM tipped with Acorn, Apple, and VLSI as its founding three — suggests that seven Founding Members is more than sufficient.

6.8 5.8 North Star Principles and Foundation Governance

When execution detail proliferates, the Foundation must retain a small number of inviolable principles that override tactical preferences. The five North Star principles are presented as the constitutional commitments of the RFL Foundation, modeled on the analogous structure derived for personal-AI platforms in Hara (2026, ch. 44):

1. **Neutrality over differentiation.** No requirement from any single vendor — Founding Member or otherwise — may be incorporated into the spec without independent demonstration of cross-vendor value.
2. **Compatibility over speed.** Backward compatibility is never broken in minor releases; new functionality enters through extensions in v1.x, never through breaking changes.
3. **Spec sovereignty absolute.** The RFL specification is released under Apache 2.0 in perpetuity. Private forks are permitted; the use of the RFL™ trademark by derivative specs is not.
4. **Layer discipline.** RFL occupies L5–L6 (embodiment abstraction + skill ISA) of the physical-AI stack. The foundation explicitly forecloses encroachment into L4 (perception models), L7 (world models), or L8 (VLA training) — the moats above and below are not RFL's to capture.
5. **Welfare over adoption.** Adoption KPIs never override welfare metrics — safety, sustainability of the developer ecosystem, fairness in voting structure. The historical lesson of Linux Foundation and RISC-V is that long-run adoption is *generated by* welfare discipline, not constrained by it.

The Foundation itself is centralized rather than federated. Centralization, properly bounded, accelerates spec evolution during the cold-start phase, attracts Founding Member commitment by offering formal voting rights, and enables clean engagement with regulatory bodies — all of which a Bitcoin-style federated structure foregoes. The risk of centralization, corruption over decades, is mitigated by constitutional immutability of the principles above, by housing the Foundation within an existing meta-foundation (the Linux Foundation is the natural candidate), and by binding the Founding Members and the executive director to public audit reporting.

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7 § 6. Implementation Roadmap

A specification proposed without a concrete maturation plan invites the suspicion that the proposal is decorative. The roadmap below is therefore presented at the level of specificity necessary to demonstrate operational seriousness, while declining the level of business detail that belongs in commercial materials rather than a technical white paper. Three trajectories run in parallel: spec versioning, reference implementation, and conformance ecosystem. A fourth — adoption milestones — emerges from their intersection.

7.1 6.1 Spec Maturation

The specification has three planned major versions on the horizon of this paper.

v0.1 (targeted for 2027 Q1). The specification described in § 4, to be frozen at release for an initial public review period of ninety days, with the companion GitHub repository containing complete JSON schemas, the BNF for the compositional algebra, and the URDF/MJCF extension. v0.1 is intended as a *review release*: the architectural commitments will be fixed at release, but the detailed schemas, error codes, and extension registry will remain open to community comment. The purpose of v0.1 is to recruit the Founding Consortium against a concrete artifact rather than against a hypothetical one.

v1.0 (targeted for 2027 Q4). The first stable specification, to incorporate Founding Consortium feedback gathered during v0.1's review period. v1.0 will be the version against which initial production deployments are certified. The version transition from v0.1 to v1.0 may modify schema details, error codes, and extension structures; from v1.0 onward, the Forward-Compatibility principle (§ 3.2 Principle 5) takes binding effect and no further breaking changes will be permitted within the v1.x line.

v2.0 (projected for 2029). The first major-version evolution. v2.0 is anticipated to incorporate emerging requirements that are visible in 2026 but not yet stable enough to specify: lifelong-learning hooks for VLAs, soft-robotics actuation extensions, and cross-modal grounding interfaces between RFL skills and ambient world-model representations. v2.0 will preserve backward compatibility with the v1.x line through a versioned extension mechanism analogous to ARM's architecture-profile evolution; v1.x-compliant implementations will continue to operate under v2.0.

A separate evolution track manages the *extension registry* — a public catalog of optional, namespaced extensions that vendors or vertical-specific working groups can propose without modifying the core spec. The extension registry process is the spec's pressure-release valve: capabilities that are too specific to belong in the core, or too novel to commit to before field validation, enter the registry and may be promoted into the core in subsequent major versions if they accumulate cross-vendor adoption. § 7 describes the governance of the registry in detail.

7.2 6.2 Reference Implementation

The reference implementation is the artifact through which the specification is exercised, demonstrated, and validated. It is open-source under Apache 2.0 and is maintained by the RFL Foundation as the canonical compliance reference.

Core (Rust). The Skill Layer parser, the Translation Layer retargeting algorithm, and the Driver Interface protocol implementation are written in Rust. The choice of language is governed by three requirements: memory safety in a system that mediates physical action, deterministic latency suitable for real-time integration, and a foreign-function interface that exposes cleanly to both Python (for ML integration) and C (for embedded firmware). Rust is the language that satisfies all three; alternatives were considered and rejected for failing on the third.

Python bindings. Generated from the Rust core through PyO3, providing the ML-research-facing API. The Python layer is designed to compose naturally with PyTorch and JAX pipelines, so that VLA developers can integrate RFL targeting with minimal friction. A LeRobot-compatible adapter ships as a v0.1 extension, allowing researchers to drive RFL-compliant embodiments through the workflow patterns they already use.

C bindings. Generated from the Rust core through cbindgen, providing the embedded-firmware-facing API. The C layer is designed to compile cleanly under the constrained toolchains common in robot controller firmware, including Yocto-based Linux distributions and the RT-thread real-time OS family.

Supported initial set (v0.1 launch, 2027 Q1). Three VLAs and three embodiments, paired in all nine combinations:

- VLAs: Physical Intelligence $\pi 0$, Stanford OpenVLA, and a Hugging Face SmolVLA variant
- Embodiments: CMU LEAP Hand, Wonik Allegro Hand, and a third embodiment of structurally distinct actuation class (e.g., a pneumatic six-finger hand), drawn from the Founding Consortium

The nine-cell support matrix is the initial demonstration that the spec is genuinely cross-vendor on both sides. It expands at v1.0 to twelve VLAs $\times$ ten embodiments and at v2.0 to the full credible deployment set.

7.3 6.3 Conformance Ecosystem

Conformance — the mechanism by which an implementation can demonstrate it meets the spec — matures through three distinct stages.

Stage 1 (targeted for 2027 Q1–Q3): self-certification. Implementers will run the open-source conformance test suite on their own hardware and publish the results. The Foundation will maintain a public registry of declared conformance results but will not independently verify them at this stage. This staging is appropriate for the early ecosystem because the cost of third-party verification is prohibitive when the deployed base is small; the integrity of the registry will rest on the reproducibility of the open-source test suite and the reputational cost to any implementer who publishes false results.

Stage 2 (targeted for 2027 Q4–2028 Q2): foundation verification. As implementer count grows, the Foundation will establish a verification capability operated by full-time engineering staff. Founding Members and other early implementers will be offered free verification as part of their membership; later implementers will pay a fee that covers the verification cost. Foundation-verified conformance will receive a distinguished badge in the public registry and will be the gate for use of the RFL™ trademark on packaging.

Stage 3 (targeted for 2028 Q3 onward): third-party notified bodies. As the conformance regime matures and the regulatory context demands independent assessment, third-party notified bodies — initially to be anchored by an MoU with TÜV or a comparable institution — will operate the verification process. This is the configuration that ultimately enables ISO 10218/13482 certification of RFL-compliant deployments to flow through the conformance result rather than through per-deployment re-evaluation, which is the operational mechanism behind the Certification reinforcement loop (§ 5.4 L4).

7.4 6.4 Adoption Milestones

The roadmap above describes what the Foundation produces. The adoption trajectory — what the ecosystem does in response — is necessarily less under the Foundation's control, but the following milestones are the indicators against which the trajectory will be evaluated.

Milestone	Target date	Indicator
Founding Consortium ratifies v0.1	2027 Q1	Seven members signed
First commercial deployment cites RFL compliance	2027 Q4	Public press release
First independent VLA targeting RFL emerges	2028 Q1	Implementation not part of Founding Consortium
First independent embodiment exposing RFL Driver Interface	2028 Q2	Same
Cross-pair integration count exceeds 50	2028 Q3	Public registry
First notified-body certification of an RFL-compliant deployment	2028 Q4	TÜV or equivalent attestation
First insurance product priced on RFL compliance	2029 Q1	Public carrier announcement

RFL referenced in a regulatory text	2029 H2	Any national or supranational AI-safety regulation

The Founding Consortium milestone is the gating event. Without it, every downstream milestone is at risk; with it, the downstream milestones become reasonable extrapolations from the consortium's commercial trajectory. § 9 of this paper is the formal invitation to the founding cohort and the operational mechanism by which the gating event is achieved.

7.5 6.5 What Is Deliberately Absent from This Roadmap

Three categories of activity are absent from the roadmap above and the absence is intentional.

Commercial product development above the spec — applications, training tools, deployment platforms — is not the Foundation's mandate and not its capability. The roadmap describes what the Foundation produces and the conformance ecosystem it stewards; it does not describe what its members or third-party vendors will build on top, because that proliferation is precisely the network-effect surface the spec is designed to enable rather than to centrally plan.

Geographic expansion is also absent. The Foundation will operate from one initial jurisdiction (the legal-entity domicile is described in § 7) and recognize compliance regimes across all jurisdictions through partnerships with relevant national bodies. There is no "EU launch" or "Asia rollout" phase because the spec, by design, is borderless.

Acquisition or merger of adjacent projects is absent. The Foundation will collaborate with LeRobot, Open X-Embodiment, ROS 2, and other adjacent efforts through technical liaison rather than corporate consolidation, in accordance with the Provider Neutrality principle (§ 3.2 Principle 4). Acquisition would compromise the perceptual independence on which the Foundation's legitimacy rests.

The roadmap is therefore narrower in scope than a commercial venture's roadmap would be, and deliberately so. The Foundation's success is measured by the breadth of the ecosystem that grows around the specification it stewards, not by the scope of activities the Foundation itself undertakes. § 7 develops the governance structure that enforces this scope discipline as a structural commitment rather than a temporary preference.

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8 § 7. Governance: The RFL Foundation

The most common reason coordination layers fail to capture the structural position their technology would otherwise warrant is governance rather than technology. Of the five adjacent efforts examined in § 2, three (Open X-Embodiment, LeRobot, NVIDIA Isaac) are foreclosed primarily by their stewardship structures rather than by any limitation of their engineering. § 7 develops the governance form designed to keep RFL from joining that list. Six elements compose the structure:

legal form, membership and voting, working groups, the specification evolution procedure, trademark and conformance policy, and the constitutional commitments that bind the Foundation in perpetuity.

8.1 7.1 Legal Form

The RFL Foundation is organized as a project under the Linux Foundation, the meta-foundation that has stewarded analogous standards efforts including OpenChain, the Cloud Native Computing Foundation, and the LF AI & Data Foundation. The Linux Foundation provides three structural advantages that an independent legal entity would have to construct from scratch: established antitrust safe-harbor protocols for multi-vendor collaboration, a tested governance template that has survived three decades of commercial pressure, and a recognized brand that signals neutrality to first-time adopters and regulators.

The Linux Foundation arrangement also provides a forcing function: the parent foundation's bylaws prohibit subsidiary projects from undertaking activities outside their declared scope, which structurally enforces the scope discipline declared in § 6.5. The trade-off accepted in choosing this form is that the Foundation operates under the Linux Foundation's revenue-allocation and policy conventions; the trade-off is judged acceptable because those conventions have been refined precisely to balance member autonomy against multi-stakeholder legitimacy.

A standalone 501(c)(6) trade association and a Swiss verein were considered as alternatives and rejected. The trade association form would require building the antitrust-protection infrastructure independently, at significant cost. The Swiss verein form, used by several standards bodies historically, would gain jurisdictional neutrality at the cost of operational complexity that early-stage standards work cannot absorb.

8.2 7.2 Membership and Voting

Membership is tiered, with three levels carrying distinct rights and obligations:

- **Founding Members** (capped at seven during the foundational period) hold the highest voting privileges and seat the initial Technical Steering Committee. Founding Member status is reserved for organizations that commit to a multi-year membership term and a documented production commitment to RFL compliance. Founding Member dues are nominal during the foundational period; the substantive obligation is engagement, not capital.
- **General Members** join after the Founding Member cohort is closed and hold voting rights on all matters except amendments to the Constitutional Commitments (§ 7.6). General Member dues are scaled to organizational revenue, with a floor that is accessible to startups (target $1,000 USD per year) and a ceiling that prevents single-organization budget dominance.
- **Community Participants** are individuals and organizations that contribute to the specification, the reference implementation, or the conformance test suite without formal membership. Community Participants do not vote but their contributions are protected by the same intellectual property and code-of-conduct policies as members'.

The seven-member cap on Founding Members is itself a constitutional commitment. It is calibrated against the historical lesson that consortia larger than approximately ten members suffer from coordination overhead that slows decision-making to the pace of the slowest member, while consortia smaller than approximately five lack the cross-vendor legitimacy that the Provider Neutrality principle requires. Seven sits at the favorable point on that curve and matches the structure that ARM's founding trio anchored before broader silicon-industry participation joined.

Voting on technical matters follows a supermajority rule (two-thirds of voting members present) for additions to the core specification, a unanimous rule for breaking changes within a major version (effectively foreclosing such changes, by design), and a simple majority for procedural matters. Founding Members and General Members vote one organization, one vote on technical matters; the structural asymmetry between tiers operates only at the governance level (board composition, treasury allocation, working group chairs).

8.3 7.3 Working Groups

The technical work of the Foundation is organized into four standing working groups, each chaired by a Founding Member representative for two-year rotating terms:

- **Specification Working Group** owns the spec document and the BNF/JSON schema artifacts. Its membership is open to all member organizations; community participation is structured through public comment periods that precede every spec change.
- **Reference Implementation Working Group** maintains the Rust core, the Python and C bindings, and the supported-embodiment matrix described in § 6.2. It operates more like a typical open-source project, with merge authority delegated to a small set of committers nominated by the working group and confirmed by the Technical Steering Committee.
- **Conformance Working Group** owns the test suite, the verification methodology, and the relationships with notified bodies. Its work is the operational basis for the Certification reinforcement loop (§ 5.4 L4).
- **Extension Registry Working Group** governs the namespaced extension catalog described in § 6.1. It is structurally distinct from the Specification Working Group to ensure that proposed extensions can be evaluated on their merits without the political pressure of immediate core inclusion.

The deliberate exclusion is a Marketing Working Group or a Business Development Working Group. The Foundation does not market itself or its members; the conformance results and the registry of compliant implementations speak for themselves, and any marketing function would create role conflicts with members' independent commercial activities.

8.4 7.4 Specification Evolution

Changes to the core specification follow a published procedure with three stages: a public Request for Comments period of ninety days, a Technical Steering Committee review with optional revision, and a final vote by the membership. The procedure applies identically to changes proposed by

Founding Members, General Members, and Community Participants, in operational instantiation of the Provider Neutrality principle.

Breaking changes within a major version are not permitted; the rule is enforced procedurally (a unanimity requirement that has historically prevented every standards body that adopted it from ratifying breaking changes) and reputationally (the Foundation's credibility rests on the forward-compatibility commitment, and any erosion of that commitment would be a foundation-level event rather than a routine governance question). Major-version transitions follow a separate, more deliberative procedure described in the Foundation bylaws.

8.5 7.5 Trademark and Conformance

The RFL™ trademark is owned by the Linux Foundation on behalf of the RFL project and is licensed only to implementations that have passed the conformance regime at Stage 2 or Stage 3 (§ 6.3). The trademark is the operational mechanism by which conformance becomes commercially legible; an implementation that claims RFL compliance without the corresponding trademark license is making a claim the Foundation can challenge.

Implementations that fail partial conformance classes may declare themselves "RFL-conformant-subset" with the specific class names, but may not use the unqualified trademark. This middle category is important: it allows research-grade implementations to participate in the ecosystem honestly without either overclaiming or being excluded.

8.6 7.6 Constitutional Commitments

Five commitments are binding on the Foundation and may be amended only by unanimous vote of the Founding Members during the foundational period and by supermajority of all members thereafter. They are the governance-layer instantiation of the principles developed in § 3.2 and recapitulated in § 5.8.

1. **Neutrality.** The Foundation will not develop, market, or invest in any VLA model, robotic embodiment, or application built on RFL. Its activity is restricted to specification stewardship, reference implementation maintenance, and conformance verification.
2. **Permissive licensing.** The specification is released under Apache 2.0 in perpetuity; the reference implementation under Apache 2.0; the conformance test suite under Apache 2.0. No future change to these licenses is permitted that would restrict downstream use.
3. **Scope discipline.** The Foundation's activities are restricted to the layer described in § 4. Expansion into adjacent layers (perception models, world models, VLA training) is structurally foreclosed.
4. **Transparency.** All Technical Steering Committee meetings are minuted and the minutes published within thirty days. The Foundation's finances are audited annually by an independent auditor and the audit report is published.
5. **Welfare primacy.** Adoption metrics never override safety, sustainability of the developer ecosystem, or fairness of the voting structure. When these come into tension, the Constitu-

tional Commitments take precedence over operational targets.

These commitments are the durable substrate of the Foundation's legitimacy. They impose costs — the neutrality commitment forecloses revenue paths that adjacent projects have used; the scope discipline forecloses expansion paths that would be tempting under commercial pressure — and the costs are accepted because they are the source of the long-term defensibility that § 5 formalizes. Coordination layers that erode their constitutional commitments under pressure do not survive; coordination layers that hold them in place across decades become the substrate on which entire industries are built.

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9 § 8. Comparative Analysis

A proposal to occupy a structural position is most credibly evaluated by examining the positions that have been occupied before and the ones that have remained empty despite attempts. This section places RFL alongside three historical coordination layers chosen for their structural relevance — ARM, CUDA, and USB — and the contemporary cohort that adjacent to RFL but does not occupy its position. The aim is not to claim RFL will replay any of these histories, but to identify the structural dimensions on which RFL must satisfy the conditions that prior successes satisfied and avoid the conditions that prior failures encountered.

9.1 8.1 The Framework

Coordination layers vary along three dimensions that matter for long-run defensibility. The first is *technical position*: the layer of the stack at which the abstraction is drawn, the side or sides of the market it abstracts, and the kind of object it standardizes (instruction, message, data format, action). The second is *governance form*: who owns the specification, who controls its evolution, how participation is regulated. The third is *constitutional discipline*: which commitments the steward refuses to abandon under commercial pressure, and which structural mechanisms enforce that refusal.

Each historical case occupies a distinct combination on these dimensions, and the combinations have proved more or less durable. The analysis below maps RFL against three of them in turn.

9.2 8.2 ARM: The Structural Correspondence

ARM is the closest structural analog and the case from which RFL's design pattern is most directly derived. § 3.3 stated the eight-element correspondence between ARM ISA design choices and RFL specification choices. § 8 emphasizes the dimensions on which ARM's history offers operational lessons for RFL's trajectory.

Three specific moments in ARM's history bear directly on choices RFL faces. First, ARM's decision in 1990 to spin out of Acorn as an independent entity — sacrificing the captive customer that funded its research to gain the neutrality required for cross-vendor adoption. The parallel for RFL is the choice to organize as a Linux Foundation subsidiary rather than as a wholly-owned project of any Founding Member, and to bind the Foundation's revenue model in ways that prevent vertical capture. The structural lesson: neutrality has a price paid up-front and a value recovered over decades.

Second, ARM's near-death experience in the desktop market from 2000–2005, when Intel's superior process technology threatened to make ARM irrelevant outside embedded niches. ARM's survival came from doubling down on the mobile market — accepting that desktop dominance was unwinnable and pivoting toward the segment where its energy-efficiency advantage compounded into a structural lead. The parallel for RFL is the imperative to choose the segment where the N$\times$M crisis is most acute and the alternative (vertical integration) is most painful — likely the industrial manipulation and humanoid-services segments where the embodiment diversity is highest and no single VLA provider has yet locked in dominant share. The structural lesson: coordination layers win the segment where vertical integration is least tolerable, not necessarily the segment of largest absolute size.

Third, ARM's response to the v8 $\rightarrow$ v9 architectural transition (2018–2022), which preserved binary compatibility for the v8 instruction base while adding new architectural features. ARM's discipline in not breaking compatibility, even when commercial parties advocated for it, protected the multi-decade application base on which its position rested. RFL's forward-compatibility principle (§ 3.2 Principle 5) is the descendant of this discipline, and § 7.4's evolution procedure is its operational instantiation.

ARM's history demonstrates that the structural position RFL seeks is occupiable, that its defensibility is durable, and that the commercial pressures against neutrality are real and recurring. ARM's history does not, however, demonstrate that physical-AI specifically will follow the same trajectory; that question is settled by execution.

9.3 8.3 CUDA: The Cautionary Inverse

CUDA occupies the cautionary inverse of ARM's position. It is a coordination layer in a technical sense — application code targeting CUDA runs across NVIDIA's GPU product line and benefits from the network of optimized libraries (cuBLAS, cuDNN, TensorRT) — but it is owned and operated by a single vendor whose primary commercial interest is selling the underlying hardware. The result is a layer of extraordinary technical sophistication that has nevertheless attracted, over the past three years, an aggressive industry-wide effort to escape it (the Triton language from OpenAI, the MAX platform from Modular, Intel's oneAPI, AMD's ROCm, Apple's Metal Performance Shaders, and the emerging cohort of inference compilers targeting non-NVIDIA accelerators).

The CUDA case shows what RFL must not become. A coordination layer whose steward also competes in adjacent markets generates a structural incentive for adopters to seek alternatives the moment the alternatives become technically viable. NVIDIA's commercial response — adding more value at the CUDA layer faster than the alternatives can erode adoption — is sustainable while NVIDIA's technical lead in silicon is unassailable, but it is not a structural defense; it is a continuously running engineering campaign. RFL's neutrality commitment (§ 7.6 commitment 1) is the structural alternative to this campaign.

The CUDA case also shows what is at stake in the choice. Had NVIDIA chosen to spin CUDA out into a neutral foundation in 2014, when its position was strong enough to dictate terms, the GPU computing ecosystem would today have a single dominant abstraction layer rather than the fragmenting cohort of alternatives that the captive ownership produced. The opportunity that NVIDIA declined is the opportunity RFL is positioned to take in physical AI before the equivalent verticalization ossifies.

9.4 8.4 USB: An Alternative Governance Model

USB occupies a third position relevant to RFL's design: it is a consortium-governed standard (the USB Implementers Forum, a non-profit), narrower in scope than either ARM or CUDA, but with notably broader cross-vendor adoption than either. The USB-IF model — a member-funded consortium that owns the specification, certifies compliant implementations, and licenses the trademark — is the closest existing governance model to the RFL Foundation form described in § 7.

Three features of USB-IF's structure are directly adopted by RFL. First, the trademark gate: USB devices cannot legally display the USB logo without passing certification, and this commercial-grade legibility of conformance has proved more effective at enforcing compatibility than any technical mechanism. RFL adopts the same pattern (§ 7.5). Second, the working-group structure: USB-IF's technical work is organized into focused working groups with rotating chairs from member organizations, which prevents single-organization dominance and ensures multiple perspectives shape each spec revision. RFL's four working groups (§ 7.3) follow the same template. Third, the modular extension mechanism: USB has evolved from 1.0 through 4.0 over thirty years by adding capabilities without breaking compatibility, governed by the same body throughout. RFL's extension registry (§ 6.1) is structurally similar.

Two features of USB-IF's structure are deliberately not adopted. USB-IF's membership dues are dominated by a small number of large hardware companies, which has produced occasional concerns about asymmetric influence over spec direction; RFL's three-tier membership structure (§ 7.2) is designed to dilute this risk. And USB-IF has no equivalent of RFL's constitutional commitments (§ 7.6); the absence has not been problematic in USB's history but is a deliberate strengthening in a domain where the long-run governance pressures are more intense.

9.5 8.5 The Contemporary Cohort

The contemporary efforts — ROS 2, LeRobot, Open X-Embodiment, and the NVIDIA Isaac/GR00T stack — were treated in § 2 as adjacent efforts that do not occupy the RFL position. From the comparative-analysis perspective, the additional observation worth making is that each occupies a different structural position from the others, and the differences themselves are evidence that the physical-AI coordination problem has been recognized but not resolved.

ROS 2 is governance-strong (Open Source Robotics Foundation, multi-stakeholder) but technical-position-narrow (transport layer, not action semantics). LeRobot is technical-position-broader but governance-weak (Hugging Face captive). Open X-Embodiment is technical-position-narrow (training data, not runtime) and governance-asymmetric (Google-dominated). NVIDIA Isaac is technical-position-broadest of the four (full stack) but governance-foreclosed (vertically integrated). Each pair of strengths is missing the complementary strength that the structural position requires. RFL is designed to combine ROS 2's governance discipline with Isaac's technical scope, neither of which exists in isolation.

9.6 8.6 The Position That Emerges

Set the analyses together and the position RFL is designed to occupy resolves to a specific combination:

Dimension	ARM	CUDA	USB	RFL
Technical layer	ISA	Compute API	Bus	**Action semantics + embodiment translation**
Sides abstracted	OS $\leftrightarrow$ silicon	App $\leftrightarrow$ GPU	Host $\leftrightarrow$ device	**VLA $\leftrightarrow$ embodiment**
Governance form	Single company (neutral)	Single vendor	Consortium	**LF-hosted foundation**
Constitutional binding	High (de facto)	None	Medium	**High (de jure)**
Verifiable safety	n/a	n/a	n/a	**Yes (§ 4.5)**
Trademark gate	License-mediated	Vendor-controlled	Certification-gated	**Certification-gated**
Cross-vendor adoption	Very high	Declining	Very high	**(target)**

The combination is novel along two axes. First, RFL pairs ARM-class technical depth with USB-class consortium governance — a combination neither historical case achieved (ARM was governance-light, USB was technically narrow). Second, RFL introduces verifiable safety semantics as a first-class architectural concern, which none of the historical cases needed to address because none of them mediated physical action.

The empty position in the historical landscape is therefore not a contingent gap but a structural one: it exists because no prior coordination problem combined the cross-vendor abstraction need with the physical-safety need at the layer where the abstraction is drawn. The physical-AI coordination problem is the first to do so, and the abstraction layer that resolves it must combine governance and verification disciplines that have not previously been combined. The Foundation, the specification, and the principles described in §§ 3–7 are the design that combines them.

10 § 9. Call to Action: The Founding Consortium

The preceding sections describe a structural opportunity. This section describes the action required to seize it and invites the seven organizations on whose participation that action depends.

10.1 9.1 Why Now

The window for a coordination layer between Vision-Language-Action models and robotic embodiments will not remain open indefinitely. § 1.4 traced the three inflections — model generality, embodiment diversity, capital intensity — whose simultaneous occurrence makes the present period decisive. § 5.3 showed that the cost of building the atomic network is bounded (approximately $50k and six months), but only if it is built before vertical integration ossifies the alternatives. § 8.3 cited NVIDIA's declined opportunity in 2014 to spin CUDA out into neutral governance, and the fragmented escape efforts that followed; the equivalent decision in physical AI must be made deliberately and now, not regretted later.

The historical pattern is consistent. ARM, USB, POSIX, and the IBM PC architecture each emerged in a window roughly eighteen months wide, after which the underlying problem had hardened into vendor-specific lock-in that no subsequent coordination effort could dislodge. We assess the equivalent window for physical AI as the period bracketing 2026 — open now, closing by approximately mid-2028.

10.2 9.2 Who We Are Inviting

We are inviting seven organizations to constitute the Founding Consortium. The composition is designed to ensure cross-vendor legitimacy on both sides of the two-sided market that RFL coordinates.

- Two to three VLA-side organizations, drawn from the cohort of foundation-model providers and prepared to commit one engineering point of contact to the Specification Working Group and one to the Reference Implementation Working Group.
- Three to four embodiment-side organizations, drawn from the cohort of robotic hand, arm, and humanoid manufacturers, and prepared to commit one engineering point of contact to the Specification Working Group and one to the Conformance Working Group.
- One academic or research-institution member, prepared to host the initial conformance test bench and contribute to the open methodology for cross-embodiment evaluation.

The seven-member cap is constitutional (§ 7.2) and will be enforced for the foundational period. The composition above is the targeted balance; the actual composition will reflect which organizations sign during the recruitment window.

10.3 9.3 What Founding Membership Entails

Founding Members commit to four obligations.

1. A multi-year membership term (initial commitment of three years) and engagement with the specification's evolution process.
2. A documented production-grade commitment to RFL compliance in at least one product line, with a defined timeline that aligns with the v1.0 release.
3. One full-time-equivalent engineering point of contact across the relevant working groups during the v0.1 $\rightarrow$ v1.0 period.
4. Adherence to the Foundation's antitrust policies, intellectual property assignment terms (Apache 2.0 outbound on contributions), and code of conduct.

Founding Members receive five rights.

1. Founding Member voting privileges on Technical Steering Committee composition and on amendments to the Constitutional Commitments (§ 7.6).
2. Free participation in the Foundation's conformance verification regime during the foundational period.
3. Priority queue access to conformance certification once the third-party notified-body regime activates (2028).
4. Use of the RFL™ trademark on compliant products without per-product royalty.
5. Public attribution as Founding Members of the RFL Foundation, in perpetuity.

The structural symmetry between Founding Members is constitutional. No Founding Member receives preferential spec influence, preferential reference implementation access, or preferential commercial terms beyond those listed.

10.4 9.4 How to Join

Organizations interested in Founding Member participation are invited to contact the project lead at the address below. The recruitment window for the v0.1 Founding cohort opens on the date of this paper's publication and closes when seven members have signed, projected for 2027 Q1.

The conversation is expected to proceed in three stages: an exploratory call to confirm strategic fit, a technical session in which the Foundation team and the prospective member's engineering leadership review the specification draft and the integration implications, and a Letter of Intent that commits both parties to the formal Founding Member agreement once the Linux Foundation subsidiary structure is operational. The full sequence is targeted at sixty days from first contact to LOI; faster paths are possible for organizations that arrive prepared.

10.5 9.5 What We Are Not Asking

It is worth stating explicitly what Founding Membership does not entail, both to address common concerns and to make the boundary of the commitment legible.

We are not asking for exclusive integration commitments. Founding Members may continue any commercial relationship, integration effort, or product roadmap they currently pursue; RFL compliance is additive, not substitutive. We are not asking for intellectual property assignment beyond the contributed code and documentation, on standard Apache 2.0 terms. We are not asking for capital investment in the Foundation; Founding Member dues during the foundational period are nominal, and the Foundation's operational funding comes from the Linux Foundation umbrella with separately budgeted grant and certification revenue. We are not asking for endorsement of any specific technical opinion held by the Foundation or its other members; the specification is the contract.

10.6 9.6 Contact

The full text of this paper, the v0.1 specification draft, the reference implementation repository, the conformance test suite, and the Founding Member application materials are maintained at the project's public website. Direct contact with the project lead can be made at the address below.

The window is open. The invitation is specific. We look forward to the seven organizations that will join us in defining the layer on which the next generation of physical AI is built.

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37. Shadow Robot Company. *Shadow Dexterous Hand (v4) Technical Specification*. 24 movements / 20 DOF tendon-driven; 20 motors; 100+ sensors at 1KHz; self-contained actuation in hand and forearm; current commercial v4 series released 2022. https://shadowrobot.com/dexterous-hand-series/ [verified 2026-05-29]

38. Bridgestone Soft Robotics Ventures (2026). *TETOTE: A Universal Robot Hand Series* (4 models: Slim, Standard, Strong-6, Strong-12). https://www.bridgestone.co.jp/products/softrobotics/tetote/ [verified 2026-05-29]

39. Tesla, Inc. (2024–2026). Optimus Hand and Arm Specifications across generations.

    - **Gen 2 reference patent**: Leddy, M. (assignee Tesla Inc.). "Underactuated hand with cable-driven fingers." WIPO Publication WO2024/073138 A1. Filed 2023-10-02; published 2024-04-04. https://patents.google.com/patent/WO2024073138A1 [verified 2026-05-30 — Google Patents]
    - **Gen 3 patent family**: Four international WIPO patents covering forearm, wrist, joint, and hand architecture; priority date 2024-10-10 (filed on the day of Tesla's "We, Robot" event); published 2026-04-16. Lead inventor on "Robotic Forearm Assembly": Konstantinos Laskaris (Tesla's former chief motor designer, now leading the Optimus program). 25 linear actuators (23 hand + 2 wrist) arranged in concentric rings around a central rotary actuator; 22-DOF hand (4 DoF $\times$ 5 fingers + 2 wrist) + 3 DoF wrist/forearm = 25 DoF per arm; 50 actuators total per robot. [partial verify 2026-05-30 — patent family count, priority date, publication date, and lead inventor confirmed; specific WIPO publication numbers per individual patent not yet indexed in public search results at verification time]
    - **Important caveat**: Per Musk disclosure (2026-04-20, Twitter/X), Tesla has "already abandoned" the Gen 3 rolling-contact design after sim-to-real failures, and the V3 hands at the summer 2026 reveal will use a newer, undisclosed design.

- Corporate announcements: Musk Gen 3 hand video disclosure 2026-02-14; specs announcement 2026-02-17.

40. Figure AI (2024). Figure unveils Figure 02, its second-generation humanoid, setting new standards in AI and robotics. *Official press release via PR Newswire*, 2024-08-06. Key specs: 16-DOF hands with human-equivalent strength, 2.25 kWh custom battery pack (50%+ energy improvement over Figure 01), 3x CPU/GPU capacity vs prior generation, 6 onboard RGB cameras. https://www.prnewswire.com/news-releases/figure-unveils-figure-02-its-second-generation-humanoid-setting-new-standards-in-ai-and-robotics-302214889.html [verified 2026-05-30]

40a. Figure AI (2025). *Introducing Figure 03*. Public technical announcement released 2025-10-09. https://www.figure.ai/news/introducing-figure-03 (Figure 02 referenced comparatively: "9% less mass and significantly less volume than Figure 02", "2x faster speeds with improved torque density".) [verified 2026-05-30]

41. Sanctuary AI (2024–2025). *Phoenix* (current public generation: **Generation 7**, launched April 2024 with improved task learning <24 hours, range of motion, weight, and bill-of-materials cost; subsequent Gen 8 data-capture variant per January 2025 announcement). Specs (Gen 7): 170 cm height, 70 kg weight, 20-DOF hands with haptic feedback, 25 kg payload, hydraulic hand actuation. https://www.sanctuary.ai/ [verified 2026-05-30 — official site lists Gen 7 as current public-product generation; Gen 8 referenced separately as data-capture release]

42. Apptronik (2025). *Apollo Humanoid Specifications*. Height 5'8" (~173cm), weight 160 lbs (~73kg), payload 55 lbs (~25kg), 4-hour battery with hot-swap. Marketed as "the first commercial humanoid robot." Target deployment: warehouses and manufacturing plants near-term; construction/oil & gas/electronics/retail/delivery/elder care future. https://apptronik.com/apollo [verified 2026-05-29]

43. Unitree Robotics. *G1 and H1 Humanoid Specifications*. G1: 1320 mm height, ~35 kg weight (with battery), 23 DOF, ~2 kg arm payload, ~2-hour battery, from $13.5K. https://www.unitree.com/g1 H1: ~180 cm height, ~47 kg, 18 DOF (4/arm, 5/leg), 3.3 m/s max speed, 189 N·m/kg peak torque density. H1-2 successor variant ~178 cm, ~70 kg, 27 DOF. https://www.unitree.com/h1 [verified 2026-05-30]

44. Seed Robotics. *RH8D Adult Robot Hand Specifications*. 19 DOF; 8 actuators; 1 kg 3D payload (2.5 kg vertical); 620 g; 9-24V; optional tactile pressure sensors (1 mN resolution). https://www.seedrobotics.com/rh8d-adult-robot-hand [verified 2026-05-29] (note: marketed as both "Inspire RH8D" and "Seed Robotics RH8D" in references; correct vendor is Seed Robotics)

11.7 G. Coordination layers in comparative analysis (§ 3, § 5, § 8)

45. Furber, S. (2000). *ARM System-on-Chip Architecture* (2nd ed.). Addison-Wesley. ISBN-13: 9780201675191. [verified 2026-05-30 — 2nd edition, 2000, Addison-Wesley publication con-

firmed]

46. ARM Limited. *Arm Architecture Reference Manual for A-profile architecture*. Current architecture revision: Armv9.7-A (2025), with copyright spanning 2020 and 2022–2025. https://developer.arm.com/documentation/ddi0487/latest [verified 2026-05-29]
47. USB Implementers Forum. *USB 4 Specification* (Version 2.0). Signaling rates 20/40/80 Gbit/s; USB-C-only connector. https://www.usb.org [verified 2026-05-29]
48. NVIDIA Corporation. *CUDA C++ Programming Guide*, Release 13.3 (2026-05-21). Features: CUDA Tile C++, CompileIQ AI-powered auto-tuning, CUDA Python 1.0. https://docs.nvidia.com/cuda/cuda-c-programming-guide/ [verified 2026-05-29]
49. The Open Group / IEEE (2024). *POSIX (IEEE Std 1003.1-2024) — Portable Operating System Interface, Base Specifications Issue 8*. Published 2024-06-14, aligned with C17 language standard. https://standards.ieee.org/ieee/1003.1/7700/ [verified 2026-05-29]

11.8 H. Structural precedents for white paper form

50. Nakamoto, S. (2008). Bitcoin: A Peer-to-Peer Electronic Cash System. (Self-published 9-page white paper, distributed via cryptography mailing list, 2008-10-31.) [Referenced for rhetorical structure, not technical content; verified 2026-05-29]
51. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems 30 (NIPS / NeurIPS 2017)*. arXiv:1706.03762. [Referenced for academic abstract register; verified 2026-05-29]

11.9 I. Industry analyst and market estimates (§ 1) — DELETED IN v1.3

Entries 52, 53, and 54 (placeholder slots for PitchBook / CB Insights VC financing aggregates, industry-analyst "suppressed market value" estimates, and 2026 robotics-AI engineering compensation surveys) were **deleted in v1.3 (2026-05-30)**. The quantitative claims those placeholders would have supported are resolved in the body via **Path A (soften)** — see `appendix_A_data_sources_en.md § A.4--A.6` for the full softening record. No body citation depends on these entries; deletion is safe.

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12 Appendices

12.1 Status

- **Appendix A**: Data Sources and Methodology — **drafted below**
- **Appendix B**: TactileManifold Formal Specification — **drafted below**
- **Appendix C**: Formal Derivations — Supermodularity and Master Equation — **drafted below**

- **Appendix D**: Author Bios — to be added once Founding team is confirmed

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13 Appendix A. Data Sources and Methodology

13.1 A.1 Provenance Status Overview

Every quantitative claim in §§ 1 and 6 falls into one of five provenance categories:

- **Verified (V)**: Sourced from a public document that has been WebFetch-confirmed in the drafting session, with the citation accurate and the figure character-identical to the source.
- **Industry-public (I)**: Sourced from a public industry report, analyst note, or company filing, but the citation has not yet been WebFetch-confirmed in the drafting session. Requires verification pass before publication.
- **Interview-pending (X)**: Figure is intended to be sourced from interviews with named or anonymized industry participants. Interviews are planned but not executed. **Publication blocker.**
- **Estimate-author (E)**: Figure is the author's estimate based on general industry knowledge, not sourced from any single citation. Acceptable in body text only if explicitly labeled as estimate.
- **Projection (P)**: Forward-looking figure derived from the model in § 5 or the roadmap in § 6. Acceptable in body text if framed as projection rather than report.

The taxonomy is enforced strictly: no claim may be presented in the body with provenance category that exceeds what is supported. The verification pass before publication consists of escalating each claim's provenance to the highest defensible category, or — if no higher provenance is available — softening the body language to match.

13.2 A.2 The Nine VLAs and Thirty Embodiments (§ 1.1)

13.2.1 VLA enumeration

Verification status (2026-05-29): Blocker #6 RESOLVED — list updated against current public sources.

The body claim of "approximately nine credible Vision-Language-Action foundation model lines" is operationalized as: models with published checkpoints or commercial APIs, demonstrated multitask generalization in third-party evaluations, and twelve-month forward maintenance commitments from an organization with public visibility. The current set, in alphabetical order:

#	Model line	Organization	Verified version (as of 2026-05-29)	Source
1	Helix	Figure AI	Active (home environments demos)	`https://figure.ai` (verified 2026-05-29)
2	Gemini Robotics	Google DeepMind	Gemini Robotics-ER 1.6 (April 2026)	`https://deepmind.google/discover/blog` (verified 2026-05-29)
3	Isaac GR00T family	NVIDIA	GR00T N1 $\rightarrow$ N1.5 $\rightarrow$ **N1.7** (current)	NVIDIA Isaac GR00T official + GitHub `https://github.com/NVIDIA/Isaac-GR00T` (verified 2026-05-29)
4	Octo	UC Berkeley	Octo-Small (27M) + Octo-Base (93M), continuing development	`https://octo-models.github.io` (verified 2026-05-29)
5	OpenVLA	Stanford / UC Berkeley / Toyota Research	7B parameter open-source release (arXiv 2406.09246), active	`https://openvla.github.io` (verified 2026-05-29)
6	π0 family	Physical Intelligence	π0 (Oct 2024, OSS Feb 2025) $\rightarrow$ π0.5 (April 2025) $\rightarrow$ π**0.7** (April 2026, current)	`https://pi.website` (verified 2026-05-29)
7	RDT-1B	Tsinghua University	1.2B parameter ICLR 2025 publication, public weights on HF	arXiv 2410.07864, HF `https://huggingface.co/robotics-diffusion-transformer/rdt-1b` (verified 2026-05-29)
8	SmolVLA	Hugging Face (LeRobot)	Multiple variants (smolvla_vlabench, robocasa, robocerebra, robomme, robotwin), ~0.5B class	`https://huggingface.co/lerobot` (verified 2026-05-29)

#	Model line	Organization	Verified version (as of 2026-05-29)	Source
9	VLA-JEPA	Hugging Face (LeRobot)	VLA-JEPA-SimplerEnv / Pretrain / LIBERO (3B class), recent release	`https://huggingface.co/lerobot` (verified 2026-05-29)

Changes from earlier draft: - **Removed**: RT-2 (superseded by Gemini Robotics line); Skild AI's foundation model (no public verification of release/maintenance status at this time) - **Added**: π0.7 (latest in PI family, April 2026); GR00T N1.5 and N1.7 progression; VLA-JEPA as a distinct Hugging Face line - **Updated**: organization attributions (OpenVLA $\rightarrow$ Stanford/UCB/TRI consortium)

Note on count: The body says "approximately nine" rather than "nine exactly" — this hedge is intentional because Hugging Face's two-line presence (SmolVLA + VLA-JEPA) could be counted as one provider or two model lines. The substantive count is 8 distinct organizations with 9 model lines, consistent with the body claim under either grouping convention.

13.2.2 Embodiment enumeration

Verification status (2026-05-29): Blocker #7 RESOLVED — list verified against current public sources across three category batches.

The body claim of "approximately thirty distinguishable embodiments in active 2026 deployment" is grounded in the following verified enumeration:

Humanoids (10 entries, verified 2026-05-29):

#	Product	Org	Verified status
H1	Optimus (Optimus 3 starting summer 2026)	Tesla	Internal testing, consumer sales targeted 2027
H2	Figure 02	Figure AI	Continuing deployment
H3	Figure 03	Figure AI	Commercially deployed at BMW Spartanburg + Leipzig
H4	NEO	1X	Consumer preorders open ($20k Early Access or $499/mo subscription, 2026 US delivery)
H5	Phoenix	Sanctuary AI	General-purpose, dexterous hand focus

#	Product	Org	Verified status
H6	Apollo	Apptronik	Active customer trials 2025-2026, $935M total Series A capital raised
H7	Digit	Agility Robotics	Enterprise-grade (~$250k+)
H8	Atlas (electric)	Boston Dynamics	Serial production began March 2026; shipping to Hyundai RMAC and Google DeepMind
H9	G1	Unitree	$16k commercial; 20,000-unit shipment target for 2026
H10	H1 / H2	Unitree	H1 ~$90k (world humanoid speed record); H2 ~$29.9k-40.9k (newly identified during this verification pass)

Plus a near-cohort from Xiaomi, ByteDance, UBTech (acknowledged in body, not enumerated here).

Multi-finger hands (10 entries, verified 2026-05-29):

#	Product	Org	Verified status
F1	RH8D Adult	Inspire / Seed Robotics	19 DOF, 8 actuators, optional tactile
F2	RH8R	Inspire / Seed Robotics	Sibling product to RH8D
F3	Allegro Hand	Wonik	4 fingers / 16 DOF, established commercial
F4	LEAP Hand	CMU (OSS)	$2k class, RSS 2023, active community
F5	Dexterous Hand v4	Shadow Robot Company	24 movements / 20 DOF, anthropomorphic, ~$100k
F6	SVH	Schunk	9 actuators, anthropomorphic, integrated wrist electronics
F7	TETOTE pneumatic series (Slim / Standard / Strong-6 / Strong-12)	Bridgestone Soft Robotics Ventures	Six-finger pneumatic hand family, payload range ~0.5–12 kg across the four models

Collaborative arms (10 entries, verified 2026-05-29 against market-leadership reports):

#	Product	Org	Verified status
A1	UR10e	Universal Robots	Market-leading cobot brand 2026
A2	UR5e	Universal Robots	Sibling SKU
A3	Panda	Franka Emika	Low-cost cobot with cloud-connected control
A4	LBR iiwa	Kuka	7-axis force-controlled, established
A5	GoFa	ABB	Single-arm cobot (~5.6% market share)
A6	YuMi	ABB	Dual-arm benchmark for micro-assembly
A7	xArm 6	UFactory	Active commercial line
A8	xArm 7	UFactory	Sibling SKU
A9	M-series	Doosan	Active commercial line
A10	Motoman HC10	Yaskawa	Active commercial line

Total: 30 verified entries. **Changes from earlier draft**: Unitree H2 (newly identified during verification, paired with H1 entry to maintain count); Boston Dynamics Atlas updated to "electric Atlas, serial production from March 2026"; Tesla Optimus annotated with "Optimus 3 production summer 2026"; Yaskawa Motoman HC10 confirmed in arm cohort.

Note on the body's "approximately thirty" framing: the count is exact at 30 in this enumeration but is presented in the body as "approximately" to allow for the rapid emergence of additional Chinese humanoid entrants (Xiaomi, ByteDance, UBTech) whose commercial-availability status as of mid-2026 is less clearly documented in English-language sources and would inflate the count to 35+ if enumerated.

13.3 A.3 Per-Pair Integration Cost (§ 1.2)

13.3.1 The $300,000 median claim

Resolution status (2026-05-29): RESOLVED — Path 2 adopted.

The body language in § 1.2 has been updated from the original draft ("a figure consistent with the integration budgets disclosed by three of the firms we interviewed during preparation of this paper") to the working-estimate framing ("based on the authors' industry experience; it is intended as a working figure rather than a precise measurement"). The interview methodology described in earlier drafts of this appendix is not planned for future revisions.

Provenance reclassification: the $300,000 median is provenance category **E (estimate-author) —** explicitly labeled as estimate in the body, with no interview backing claimed and none planned.

Component cost ranges cited in § 1.2 — direct labor ($50k–$180k), hardware procurement ($20k–$100k), data collection ($30k–$200k), certification ($50k–$300k) — remain author estimates as originally drafted. They stand on their own under the explicit estimate framing introduced by this resolution.

Implication for downstream claims: the aggregate fragmentation tax figure in § 1.3 ($30M direct, derived from "100 active pairs $\times$ $300k median") inherits the estimate framing. The body should not be read as asserting an empirically validated figure; readers seeking empirical validation are directed to this resolution note.

Decision rationale (for the record): the alternative (Path 1: conduct three structured interviews) would have delayed publication by an estimated 30–45 days. The judgment was that the persuasive marginal value of interview-backed framing did not justify the delay, given that (a) the body's other arguments (§ 5 network effects, § 8 comparative analysis) carry the paper's strategic claim independently of the $300k figure's precision, and (b) the estimate framing is honest and survives academic scrutiny on its own.

13.3.2 The $25–30k engineer-month compensation claim

Resolution status (2026-05-29): RESOLVED — Path A' adopted.

The original body language ("$25,000–$30,000 per month fully loaded for senior robotics-AI talent") has been replaced with directional framing: "high tens of thousands of dollars per engineer-month in the dominant US labor market — consistent with senior-IC bands at leading AI laboratories." The downstream per-pair direct labor cost derivation ($50,000–$180,000) has been correspondingly softened to "low tens of thousands of dollars at the lower bound to the low hundreds of thousands at the upper bound."

Provenance reclassification: from I (industry-public, source-required) to E (estimate-author, directional). No specific compensation survey is cited; readers may consult public sources (Levels.fyi, Robert Half) for empirical anchoring.

Decision rationale: consistent with the soften principle adopted for Blockers #1 and #2. The compensation figure is publicly verifiable through several sources, so citation would have been feasible (unlike Blockers #1 and #2). The choice to soften rather than cite reflects the project's blanket Path-2/Path-A resolution policy: avoid creating verification dependencies in body claims where directional language is sufficient for the strategic argument. Readers wanting precise figures are directed to the public compensation databases listed.

13.4 A.4 Aggregate Fragmentation Tax (§ 1.3)

The body's aggregate fragmentation tax is constructed as the sum of four components. Status updated 2026-05-29 (Blocker #2 RESOLVED via Path A — soften):

Component	Body figure (post-resolution)	Provenance	Derivation
Direct integration spend	$30M+/year	E (estimate)	100 active pairs × $300k median — derived, not measured
Maintenance overhead	$6–9M/year	E	20-30% of $30M direct — author estimate of typical software-maintenance ratio
Capability suppression	“high hundreds of millions to low billions, no point estimate”	E (estimate, directional)	Original $500M–$2B range softened to directional language per Path A
Lock-in penalty	unquantified	P (projection)	Compounding model, not yet quantified

Resolution note (2026-05-29): the original “$500M–$2B” range carried provenance category X (interview/analyst-pending). Following the same decision principle adopted for Blocker #1, Path A (soften the body language) was chosen over Path B (source from a specific analyst report). The component is now expressed as a directional estimate without a point figure, and the headline aggregate is reformulated correspondingly.

Headline aggregate (post-resolution): “plausibly on the order of low single-digit billions in 2026” (was “$1–3 billion in 2026”), with the explicit note that the figure inherits the uncertainty of the dominant suppression component. The change is propagated to the Abstract for consistency.

The “low tens of billions by 2030” projection is provenance category **P** (projection from the trajectory model in § 5.5 + § 6.4). This is acceptable in the body as long as it is framed as projection rather than report.

13.5 A.5 Capital Aggregates (§ 1.4)

Resolution status (2026-05-29): RESOLVED — Path A adopted for both Blockers #4 and #5 (batched).

The original body language (“over $30 billion in disclosed venture financing” and “more than $15 billion” humanoid subset) has been softened to “disclosed venture financing in the tens of billions of dollars, with humanoid robotics accounting for a substantial fraction — by independent aggregator estimates, on the order of half the total.”

Provenance reclassification: both figures move from I (industry-public, source-required) to E (estimate-author, directional). The body retains the strategic point (capital is committed at unprecedented scale; humanoid subset is significant) without committing to point estimates that would require verification against PitchBook, CB Insights, or Crunchbase aggregates.

Decision rationale: consistent with the project-wide soften policy (Blockers #1–#3). Sources to consult if a future revision elects to re-introduce point estimates: PitchBook Robotics Funding Quarterly, CB Insights State of Robotics, ARK Invest Big Ideas reports, Crunchbase aggregate exports. No commitment is made to do so.

13.6 A.6 Roadmap Milestone Dates (§ 6)

Verification status (2026-05-29): Blocker #8 RESOLVED — future-tense framing made explicit throughout § 6.

The dates in § 6.1 (v0.1 targeted for 2027 Q1, v1.0 targeted for 2027 Q4, v2.0 projected for 2029) and § 6.4 (Founding Consortium ratification 2027 Q1, etc.) are provenance category **P** (projection). They are derived from the 18-month execution plan in [[STRATEGY.md § 9]] and the cold-start dynamics in § 5.3 / § 5.4. They are acceptable in the body as projections.

Body language calibration (RESOLVED 2026-05-29): § 6 has been audited and tightened for explicit future-tense framing. Specific changes: v0.1/v1.0/v2.0 headers now use "targeted for"/"projected for" prefix; Stage 1/2/3 conformance descriptions in § 6.3 converted from narrative present tense ("Implementers run", "Foundation establishes") to explicit future tense ("Implementers will run", "Foundation will establish"); v1.0 and v2.0 descriptive sentences propagated will/will-not auxiliaries. § 6.4 table column "Target date" was already projection-language compliant. The audit confirms no remaining date references imply commitment beyond targeting.

13.7 A.7 Publication-Blocker Summary

Eight specific claims must be resolved before public release. Status updated as resolution proceeds.

1. ~~**§ 1.2 "three firms we interviewed**~~ **— RESOLVED 2026-05-29** (Path 2: relabeled as author estimate; see A.3 above)
2. ~~**§ 1.3 "$500M–$2 suppressed market value**~~ **— RESOLVED 2026-05-29** (Path A: softened to "high hundreds of millions to low billions" directional; headline aggregate softened to "low single-digit billions"; Abstract updated for consistency; see A.4 above)
3. ~~**§ 1.2 "$25–30 engineer-month compensation**~~ **— RESOLVED 2026-05-29** (Path A': softened to "high tens of thousands of dollars per engineer-month"; downstream $50–180k derivation correspondingly softened; see A.3 above)
4. ~~**§ 1.4 "$30B disclosed VC financing**~~ **— RESOLVED 2026-05-29** (Path A, batched with #5: softened to "tens of billions of dollars"; see A.5 above)
5. ~~**§ 1.4 "$15B humanoid subset**~~ **— RESOLVED 2026-05-29** (Path A, batched with #4: softened to "on the order of half the total"; see A.5 above)
6. ~~**§ 1.1 "nine credible VLAs**~~ **— RESOLVED 2026-05-29** (Path B: each entry WebFetch-verified; list updated to current model lines including π0.7, GR00T N1.7, VLA-JEPA; RT-2 and Skild AI removed; "nine" softened to "approximately nine"; see A.2 above)
7. ~~**§ 1.1 "thirty distinguishable embodiments**~~ **— RESOLVED 2026-05-29** (Path B: 30 entries WebFetch/WebSearch-verified across 3 category batches; Unitree H2 added, Boston Dynam-

ics electric Atlas annotated, Tesla Optimus 3 dated; see A.2 above)

8. **~~§ 6 dates~~ — RESOLVED 2026-05-29** (audit pass: v0.1/v1.0/v2.0 headers prefixed with "targeted for"/"projected for"; Stage 1/2/3 narrative converted to explicit future tense; see A.6 above)

Progress: 8 of 8 resolved. ✓ All publication blockers from this list closed.

13.8 A.8 Confidence Levels for Headline Claims

Body claim	Current confidence	Confidence after verification pass
"$1–3 billion annually and rising" (fragmentation tax)	Low — depends on unsourced component	Medium — once analyst figure is sourced
"$300,000 median per pair"	Low — interview-pending	High — once interviews complete
"270 candidate integration pairs (9×30)"	High — derived arithmetic	High (unchanged)
"$30B / $15B VC financing 2024–2025"	Medium — industry-knowledge	High — once aggregate is verified
"approximately eighteen months window"	Medium — author judgment based on § 1.4 inflections	Medium (unchanged; this is necessarily a judgment)

13.9 A.9 Interview Protocol (DEPRECATED 2026-05-29)

This section originally specified the interview protocol that would resolve § 1.2's publication blocker via Path 1. Following the Path 2 resolution adopted on 2026-05-29 (see A.3), interview-based validation is **not planned** for this paper or its revisions. The section is retained in deprecated form for historical traceability of the resolution decision.

(Original protocol — superseded; not to be executed unless the resolution decision is revisited.)

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14 Appendix B. TactileManifold: Formal Specification

The TactileManifold is the RFL Translation Layer's canonical representation of tactile state. Its design satisfies a constraint that is unmet by any existing tactile-sensor format: it must express the observations of a binary four-channel contact sensor, a three-axis force-torque sensor, a high-resolution GelSight-class visuotactile sensor, and as-yet-undesigned tactile modalities in a single uniform structure that admits capability negotiation. This appendix gives the formal definition, the resolution hierarchy that enables negotiation, and the realizations for the principal sensor classes in 2026 commercial use.

14.1 B.1 Motivation and Design Constraints

The body section establishes the structural requirement: a VLA expressing a high-level skill (e.g., `grasp_pinch`) must be able to specify a tactile target without committing to a particular sensor format. An embodiment receiving the target must be able to report its observations against it. Different embodiments with different sensors must produce comparable observations. The abstraction therefore satisfies three constraints simultaneously:

1. **Format independence**: a sensor's native output format does not appear in the manifold definition. Sensor-specific encoding is performed by the embodiment's Driver Interface during translation.
2. **Resolution heterogeneity**: a low-resolution sensor and a high-resolution sensor must express observations in the same abstraction, with the resolution difference captured as a parameter rather than as a different structure.
3. **Graceful degradation**: if a VLA requests a feature the embodiment cannot supply, the negotiation must fail cleanly with a structured error, not silently produce a degraded observation that the VLA misinterprets.

The third constraint is the load-bearing one. Many existing tactile-fusion approaches produce a single output that hides the underlying resolution differences, with the consequence that a VLA trained on high-resolution data degrades unpredictably when deployed against a low-resolution sensor. The TactileManifold abstraction is designed to make the resolution explicit at every layer.

14.2 B.2 Formal Definition

A TactileManifold is a tuple

$$\mathcal{M} = (S, \mathcal{F}, R, t)$$

where:

- $S \subseteq \mathbb{R}^3$ is a *spatial domain*, a (possibly discrete) subset of the embodiment's end-effector surface, equipped with a topology induced by the surface metric. For dense sensors, S is a two-dimensional manifold patch; for discrete sensors, S is a finite set of points; both are admitted by the definition.
- $\mathcal{F} : S \to \mathbb{V}$ is a *feature field*, assigning to each point in S a vector of contact features drawn from a *feature vocabulary* $\mathbb{V}$. The vocabulary is defined below and is the structure that admits cross-modality comparison.
- $R = (r_s, r_t, r_f)$ is a *resolution triple*, where r_s is the spatial density (points per unit area, or "discrete" for sparse sensors), r_t is the temporal sample rate (Hz), and $r_f : \mathbb{V} \to \mathbb{N}$ is a per-feature precision function (bits of effective dynamic range per feature channel).
- $t \in \mathbb{R}_{\geq 0}$ is the timestamp of the manifold instance.

The feature vocabulary $\mathbb{V}$ is defined as a fixed, extensible set of named feature channels:

$$\mathbb{V} = \{\text{contact}, \text{normal\_force}, \text{shear}_x, \text{shear}_y, \text{slip}, \text{vibration}, \text{temperature}, \text{compliance}, \ldots\}$$

Each named channel has a fixed semantic interpretation specified in the RFL spec document. The set is extensible through the extension registry (§ 6.1), so future sensor modalities can register new channels without breaking the core specification.

A *partial manifold* is an instance in which $\mathcal{F}$ is defined only over a subset of the vocabulary. The spec language requires the embodiment's capability manifest to declare which channels it supplies, so the partial nature is contractual and verifiable rather than silent.

14.3 B.3 The Resolution Hierarchy

The resolution triple R admits a partial order $\succeq$:

$$R_1 \succeq R_2 \iff r_{s,1} \geq r_{s,2} \ \wedge \ r_{t,1} \geq r_{t,2} \ \wedge \ r_{f,1}(c) \geq r_{f,2}(c) \ \forall c \in \mathbb{V}$$

This partial order is the structure that supports capability negotiation. When a VLA emits a Skill Layer invocation with a tactile target specified at resolution R_{req}, the Translation Layer queries the embodiment's capability manifest for the native resolution R_{native}. If $R_{\text{native}} \succeq R_{\text{req}}$, the invocation proceeds; the embodiment supplies a manifold instance at R_{native} (or, optionally, downsamples to R_{req} to reduce bandwidth). If $R_{\text{native}} \not\succeq R_{\text{req}}$, the negotiation fails with a structured error identifying the dimension(s) on which the requested resolution exceeds the embodiment's capability.

The asymmetric nature of the partial order is intentional. A VLA that requests low resolution and receives high resolution can use the additional information or discard it without error. A VLA that requests high resolution and receives low resolution cannot infer the missing information; the failure must be visible. The Translation Layer enforces this by treating the resolution declaration as a contract rather than a hint.

14.4 B.4 Realizations from Specific Sensor Classes

Five sensor classes in 2026 commercial use are realized as follows. The mapping shown for each is the canonical one defined in the RFL Reference Implementation; embodiment vendors may register alternative mappings through the extension registry if their sensor admits more refined representation.

Binary contact sensors (four-channel discrete). Used in many low-cost grippers. Mapping: S is a discrete four-point set corresponding to fingertip locations; $\mathcal{F}(s) = \{\text{contact: } \{0, 1\}\}$ for each $s \in S$; $R = (\text{discrete}, 100\,\text{Hz}, r_f(\text{contact}) = 1)$. Only the contact channel is populated; the remaining vocabulary is undefined.

Three-axis force-torque sensors at the wrist. Used in most collaborative arms. Mapping: S is a single-point set at the force-sensor location; $\mathcal{F}(s) = \{\text{normal\_force}, \text{shear}_x, \text{shear}_y\}$ as three scalars; $R = (\text{discrete}, 1000\,\text{Hz}, r_f(\cdot) = 16)$. The three force channels are populated; spatial channels are absent because the sensor is point-localized.

Distributed force-sensor arrays (XELA-class). Used in high-end multi-finger hands. Mapping: S is a discrete set of sensor locations (typically 8–24 per fingertip); $\mathcal{F}(s) = \{\text{normal\_force}, \text{shear}_x, \text{shear}_y\}$ at each location; $R = (\text{discrete}_n, 100\,\text{Hz}, r_f(\cdot) = 14)$ where n is the per-fingertip array size.

Visuotactile sensors (GelSight, DIGIT classes). Used in research-grade and emerging industrial deployments. Mapping: S is a dense two-dimensional patch (e.g., 320 × 240 pixels mapped to the elastomer-gel surface); $\mathcal{F}(s) = \{\text{normal\_force}, \text{shear}_x, \text{shear}_y, \text{slip}\}$ derived from the gel-deformation image; $R = (\text{dense}, 30\,\text{Hz}, r_f(\cdot) = 8)$.

Pneumatic pressure sensors (as in McKibben and bellows-style pneumatic hands). Used in pneumatic-hand embodiments. Mapping: S is a discrete set of finger-chamber locations (e.g., six chambers in a six-finger pneumatic hand); $\mathcal{F}(s) = \{\text{normal\_force derived from chamber pressure}\}$; $R = (\text{discrete}, 50\,\text{Hz}, r_f(\text{normal\_force}) = 12)$. The pressure-to-force conversion is calibrated through the embodiment descriptor's calibration metadata.

This list is not exhaustive; the design intent is that any tactile sensor whose output can be characterized by a spatial distribution of feature values can be expressed in TactileManifold form. The capability negotiation mechanism then handles the heterogeneity at runtime.

14.5 B.5 Capability Negotiation Semantics

For a Skill Layer primitive whose tactile target is $\mathcal{M}_{\text{req}} = (S_{\text{req}}, \mathcal{F}_{\text{req}}, R_{\text{req}}, \cdot)$, the negotiation against an embodiment with native capability $\mathcal{M}_{\text{native}} = (S_{\text{native}}, \mathcal{F}_{\text{native}}, R_{\text{native}}, \cdot)$ proceeds in three steps:

1. **Feature coverage**: every channel in $\mathcal{F}_{\text{req}}$ must be present in $\mathcal{F}_{\text{native}}$. If not, negotiation fails with `UnsupportedFeatureError(channels=...)`.
2. **Spatial coverage**: S_{req} must be approximable by points in S_{native} within a declared spatial tolerance. If not, negotiation fails with `SpatialCoverageError(uncovered_region=...)`.
3. **Resolution adequacy**: $R_{\text{native}} \succeq R_{\text{req}}$ as defined in B.3. If not, negotiation fails with `InsufficientResolutionError(dimensions=...)`.

The Driver Interface returns the structured error rather than silently substituting a fallback. The VLA's response to negotiation failure is its own concern; the Translation Layer's responsibility is to make the failure visible and structured.

The negotiation produces, on success, an *adapted manifold* $\mathcal{M}_{\text{adapted}}$ which is the embodiment-native manifold downsampled to R_{req} (if downsampling is requested) or supplied at R_{native} (the default).

The VLA receives observations against $\mathcal{M}_{\text{adapted}}$, with a metadata field indicating which resolution was used.

14.6 B.6 Open Questions

Two questions are left to future spec revisions.

First, the formal treatment of *deformable-object tactile interaction* — where the contact surface itself deforms during interaction and the spatial domain S should be time-varying — is sketched but not specified. The current spec accepts S as time-invariant during a single primitive invocation; future revisions will need to address adaptive spatial domains for cloth, soft-object, and slip-rich manipulations.

Second, the relationship between TactileManifold and the *proprioceptive* state captured separately in the canonical action representation (§ 4.2) admits some overlap (force at the wrist appears in both). The current spec treats them as separate channels; a future revision may unify them under a richer end-effector-state abstraction.

These are flagged for v2.0 of the specification, consistent with the extension-registry policy (§ 6.1) by which the v0.1 / v1.0 core stays small and well-defined while non-trivial new structure migrates through extensions before promotion.

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15 Appendix C. Formal Derivations

This appendix formalizes two propositions stated informally in § 5: the supermodularity of bilateral translation value (§ 5.2) and the multiplicative-with-constraints structure of the RFL value function (§ 5.5). The arguments are mathematically straightforward once the setup is precise; the appendix exists primarily to make the setup precise.

15.1 C.1 Notation and Setup

Let $\mathcal{V}$ denote the (countable) set of Vision-Language-Action models in active use, indexed by $v \in \mathcal{V}$, and let $\mathcal{E}$ denote the (countable) set of robotic embodiments, indexed by $e \in \mathcal{E}$. A *deployment instance* is a pair $(v, e) \in \mathcal{V} \times \mathcal{E}$ paired with an environment context $\xi \in \Xi$ and a task specification $\tau \in \mathcal{T}$.

For each deployment instance, the RFL Translation Layer realizes a function

$$T : \mathcal{A}_v \times \Theta_v \times \Phi_e \to \mathcal{A}_e$$

where $\mathcal{A}_v$ is the action space output by VLA v, Θ_v is the information held by the Translation Layer about v's semantics (its action distribution, its calibration, its uncertainty model), Φ_e is the information held about embodiment e (its kinematic structure, its actuation modality, its safety envelope), and $\mathcal{A}_e$ is the embodiment-specific command space.

Define the *translation value* $V(\theta, \phi)$ as the expected utility of the translation across a representative distribution over (ξ, τ):

$$V(\theta, \phi) = \mathbb{E}_{(\xi,\tau)} \left[U(T(a; \theta, \phi)) \mid a \sim P_v(\cdot \mid \xi, \tau) \right]$$

where $\theta \in \Theta_v$ is the VLA-side information level, $\phi \in \Phi_e$ is the embodiment-side information level, a is the action sampled from the VLA, and U is a scalar utility composing task success, safety-envelope adherence, and execution efficiency.

We treat θ and ϕ as scalar information levels (e.g., empirical KL-divergence between the Translation Layer's belief over the relevant distribution and the true distribution). The vector-valued generalization is straightforward and is omitted to keep notation legible.

15.2 C.2 Supermodularity of Translation Value

Proposition 1 (Supermodularity of V). Under regularity conditions on U and the underlying distributions stated below, the value function $V : \Theta_v \times \Phi_e \to \mathbb{R}$ is supermodular in (θ, ϕ). Equivalently, for any $\theta_L < \theta_H$ and $\phi_L < \phi_H$:

$$V(\theta_H, \phi_H) - V(\theta_H, \phi_L) \geq V(\theta_L, \phi_H) - V(\theta_L, \phi_L) \tag{C.1}$$

with strict inequality whenever both $\theta_H - \theta_L > 0$ and $\phi_H - \phi_L > 0$ have non-trivial impact on T's output.

Regularity conditions.

1. U is differentiable and bounded on the relevant compact set.
2. The map T is jointly continuous in (a, θ, ϕ) and differentiable almost everywhere.
3. The distributions P_v and the embodiment-side noise distributions are not degenerate (i.e., the bilateral information actually matters; neither side is fully observable from the other).

Proof sketch. The argument follows Topkis (1998) Theorem 2.6.1, adapted to the translation setting.

Step 1: Cross-partial sign. For V twice continuously differentiable in (θ, ϕ), supermodularity is equivalent to $\partial^2 V / \partial\theta\, \partial\phi \geq 0$ everywhere on the domain (Topkis 1998, Lemma 2.6.2). We compute this cross-partial.

Holding a fixed, the contribution to V at point (θ, ϕ) from any single a is $u(T(a; \theta, \phi))$ where $u = U$ composed with the post-translation execution model. The cross-partial of $u(T(a; \theta, \phi))$ in (θ, ϕ) is:

$$\frac{\partial^2 u(T)}{\partial\theta\, \partial\phi} = u''(T) \cdot \frac{\partial T}{\partial\theta} \cdot \frac{\partial T}{\partial\phi} + u'(T) \cdot \frac{\partial^2 T}{\partial\theta\, \partial\phi}$$

Group the two terms by mechanism. The **complementarity channel** is the second term, $u'(T) \cdot \partial^2 T/\partial\theta\,\partial\phi$, and it is non-negative: marginal utility is positive ($u' > 0$), and the cross-partial of the translation outcome is itself non-negative under the structural assumption that VLA-side and embodiment-side information *combine constructively*. This is the substantive claim. It follows from the form of the optimal translation: under joint information (θ_H, ϕ_H) the translator exploits the joint distribution, whereas under either piece alone it must marginalize over the missing dimension, which injects irreducible variance; the variance removed by adding one side's information is larger when the other side's information is already present, so $\partial^2 T/\partial\theta\,\partial\phi \geq 0$.

The **curvature channel** is the first term, $u''(T) \cdot (\partial T/\partial\theta)(\partial T/\partial\phi)$. The two single-side partial derivatives share sign (improving either side moves T in the same direction, toward the true optimal command), so their product is non-negative; under a concave, risk-averse valuation ($u'' \leq 0$) this term is therefore *non-positive* — curvature alone induces a mild substitution between the two information sources — and it vanishes for a risk-neutral (locally linear) valuation.

The cross-partial is therefore non-negative, and V is supermodular, in either of two regimes: (i) **risk-neutral valuation** ($u'' \equiv 0$), where $\partial^2 V/\partial\theta\,\partial\phi = \mathbb{E}_a[\partial^2 T/\partial\theta\,\partial\phi] \geq 0$ holds unconditionally; or (ii) **risk-averse valuation**, where the complementarity channel dominates the curvature substitution,

$$u'(T)\,\frac{\partial^2 T}{\partial\theta\,\partial\phi} \;\geq\; |u''(T)|\,\frac{\partial T}{\partial\theta}\,\frac{\partial T}{\partial\phi}.$$

This dominance holds in the regime the marginalization argument identifies — where joint information removes first-order outcome variance that neither side removes alone — because the complementarity channel is then first-order in the recovered information while the curvature substitution is second-order in the single-side gradients.

Step 2: Integration over a. Taking the expectation over $a \sim P_v$ preserves the sign of the cross-partial pointwise (by linearity of expectation), giving $\partial^2 V/\partial\theta\,\partial\phi \geq 0$ on the domain under the Step-1 regime (risk-neutral valuation, or the dominance condition under risk aversion), with strict inequality wherever the constructive-combination assumption is non-degenerate.

Step 3: Supermodularity inequality. Integration of the cross-partial twice (once in θ, once in ϕ) over the rectangle $[\theta_L, \theta_H] \times [\phi_L, \phi_H]$ yields the supermodularity inequality (C.1). ■

Connection to Crémer-McLean (1985, 1988). The original Crémer-McLean result, in the mechanism design setting, establishes that a monopolist with knowledge of the correlation structure between bidders' types can extract full surplus, while a monopolist with knowledge of only the marginal distributions cannot. The translation setting is structurally analogous: the Translation Layer is a "mechanism" whose payoff depends on the joint structure of two distributions (VLA action distribution and embodiment response distribution), and bilateral information enables strict outcome dominance over unilateral information. This is a structural analogy, not an application of the theorem. The Crémer-McLean full-surplus-extraction result rests on strong and famously non-robust assumptions — correlated types, a commonly known correlation structure, and unbounded transfers — that the translation setting does not claim to satisfy; it is cited only to locate the bilateral-information complementarity within a recognized lineage in mechanism design. The load-bearing formal claim is the Topkis-form supermodularity of Steps 1–3, which stands independently of Crémer-McLean.

Empirical signals. The proposition's empirical consequence — that joint exposure to multi-VLA $\times$ multi-embodiment data outperforms either unilateral case — is directionally consistent with published results. Physical Intelligence (2024) reports improvements when π_0 is fine-tuned on multi-embodiment data relative to single-embodiment fine-tunes of the same parameter count; whether that gain is strictly super-linear (as the proposition predicts) rather than merely positive is not established by the cited results, and we do not claim it here. The Open X-Embodiment results (Padalkar, Pooley, Jain, et al., 2023) show analogous transfer gains across multiple embodiments. Two cautions are warranted. First, these results are consistent with the proposition but do not, in themselves, prove it. Second, multi-embodiment *training* gains are equally consistent with end-to-end models that internalize the cross-embodiment structure directly: the proposition motivates the bilateral-information complementarity, but it does not establish that an explicit external translation layer is the only — or the best — way to realize that complementarity. That case rests on the coordination, neutrality, and conformance arguments of the body, not on this proposition alone.

15.3 C.3 The RFL Master Equation

We turn now to the Master Equation stated in § 5.5:

$$\text{Value}(t) = \int_0^t \beta \cdot N_{\text{VLA}}(s) \cdot N_{\text{emb}}(s) \cdot |\Sigma(s)| \cdot Q_{\text{tr}}(s) \cdot T_{\text{cert}}(s)\, ds \tag{C.2}$$

subject to constraint set $\mathcal{C}$ given in § 5.5. This appendix derives the equation's functional form from underlying network-effect dynamics and shows that the constraint set is *binding* in the formal sense (violation of any constraint drives $\text{Value}(t) \to 0$).

15.3.1 C.3.1 The Network-Effect Integrand

Consider the *instantaneous* value of the RFL coordination layer at time s. By the indirect-network-effect structure of the two-sided market formalized in § 5.1, the instantaneous value scales with the *number of cross-side combinations* that the layer enables. With $N_{\text{VLA}}(s)$ compliant VLAs and $N_{\text{emb}}(s)$ compliant embodiments, the gross combination count is $N_{\text{VLA}}(s) \cdot N_{\text{emb}}(s)$ — the avoided per-pair integration burden that § 1.1 quantifies.

The combination count is not, however, sufficient. Each combination is only as valuable as the *quality* of the translation between its endpoints and the *trust* the surrounding ecosystem places in that translation. The skill library size $|\Sigma(s)|$ scales the *expressive surface* of the combinations (a combination is valuable in proportion to the number of skills available to it); the translation quality

$Q_{\text{tr}}(s) \in [0,1]$ scales the *fidelity* with which any given skill executes; and the cumulative certification trust $T_{\text{cert}}(s) \in [0,1]$ scales the *deployable surface* (a combination is only commercially deployable to the extent it is trusted, which Section 6.3's three-stage conformance regime instantiates).

These five factors enter multiplicatively rather than additively because each is, by the structural argument in § 5.4, a *necessary* condition for instantaneous value, not an *additive contributor* to it. A coordination layer with high VLA count but zero embodiment count produces zero value (nothing to combine); a high skill library with zero translation quality produces zero value (the library cannot be executed); high translation quality with zero trust produces zero deployment. The multiplicative form is the formal expression of joint necessity.

The scaling coefficient β absorbs the units and the calibration constants. Its dimensional analysis: β has units of [value] / [VLA count $\times$ embodiment count $\times$ skill count $\times$ time]. Empirical calibration is discussed in C.3.5.

15.3.2 C.3.2 Multiplicative versus Additive Compounding

The instantaneous integrand $\beta \cdot N_{\text{VLA}} \cdot N_{\text{emb}} \cdot |\Sigma| \cdot Q_{\text{tr}} \cdot T_{\text{cert}}$ scales as the *product* of its factors. We contrast this with the *sum* of its factors to make the difference precise.

Property 1 (Multiplicative gain). If each of five factors doubles, the integrand grows by $2^5 = 32\times$. The additive analog $(2N_{\text{VLA}} + 2N_{\text{emb}} + 2|\Sigma| + 2Q_{\text{tr}} + 2T_{\text{cert}})/5$ grows by only $2\times$.

Property 2 (Multiplicative collapse). If any single factor drops to zero, the integrand collapses to zero. The additive analog merely reduces by one-fifth.

The multiplicative form therefore exhibits both the upside compounding that the body section celebrates and the downside fragility that demands the constraint set. ARM's silicon-side market did not exhibit this fragility because no analog of T_{cert} existed (silicon faces no comparable certification regime). RFL's physical-action substrate makes the multiplicative form structurally appropriate.

15.3.3 C.3.3 The Constraint Set as Binding

The Master Equation is subject to five constraints, repeated here from § 5.5 with notational consistency:

$$
\begin{aligned}
\text{Neutrality}(s) &\geq 0.9 && \text{(no single vendor exceeds 30\% governance share)} \\
\text{SpecStability}(s) &\geq 0.95 && \text{(backward-compatibility ratio)} \\
\text{IntegrationCost}(s) &\leq \$50{,}000 && \text{(per (VLA, embodiment) pair)} \\
\text{Safety}(s) &= 1 && \text{(ISO 10218/13482 compliance)} \\
\text{GovTransparency}(s) &= 1 && \text{(Foundation meetings public)}
\end{aligned}
$$

Proposition 2 (Binding nature of $\mathcal{C}$). Violation of any single constraint at time $s^* \in [0,t]$ drives $T_{\text{cert}}(s) \to 0$ for all $s > s^*$.

Sketch of justification. Each constraint corresponds to a precondition for one of the reinforcement loops L4 through L9 identified in § 5.4. Loss of neutrality collapses the cross-vendor adoption that L4's certification regime requires (Founding Members defect). Loss of spec stability collapses the developer investment that L3 requires (developers cannot rely on the layer). Excess integration cost collapses the marginal-pair adoption that drives L1's data loop. Loss of safety compliance collapses L8 (insurance) and L9 (regulatory) immediately and reflexively erodes L4. Loss of governance transparency collapses the institutional trust that L4 — L9 all require.

Because $T_{\text{cert}}(s)$ is a multiplicative factor in the integrand, its collapse to zero halts further value accumulation. This is the formal expression of the body's claim that the constraints are *binding*: they are not soft preferences but architectural necessities. ■

15.3.4 C.3.4 Path Dependence

The integral form of (C.2) captures path dependence: the value at time t depends on the trajectory of the factors over $[0, t]$, not merely on their values at t. This is consequential for the cold-start period. Two trajectories that arrive at the same factor values at time t by different paths produce different cumulative values:

$$\text{Value}_{\text{slow-start}}(t) < \text{Value}_{\text{fast-start}}(t)$$

even when both reach $(N^*_{\text{VLA}}, N^*_{\text{emb}}, |\Sigma|^*, Q^*_{\text{tr}}, T^*_{\text{cert}})$ at t. The slow-start trajectory accumulates the integrand at lower intermediate values for longer; the fast-start trajectory accumulates at near-final values for longer.

This is the formal basis for § 5.3's atomic-network urgency argument: late entry into a coordination position that requires multi-year accumulation does not merely delay value; it foregoes value that cannot be recovered. Late entrants must invest more aggressively to *catch up* in the integrand, against an incumbent who continues to accumulate at high values throughout. The historical analog is ARM's recovery of Intel's intervening years in the mobile market — possible only because ARM had been accumulating mobile-side value throughout the period during which Intel was capturing the desktop integral.

15.3.5 C.3.5 Calibration Notes

The scaling coefficient β is a free parameter in (C.2). For projection purposes, calibration against historical platform-value growth curves is feasible if not exact. ARM's value trajectory from 1990 (founding) to 2016 (Softbank acquisition at $32B) provides one calibration anchor; CUDA's revenue trajectory from 2007 to 2024 provides a second. Both anchors suggest β values in the range that would produce, for RFL, low-billions-of-dollars cumulative value by 2032 conditional on the trajectory targets in § 6.4 being met. We treat these as illustrative rather than predictive; precise β calibration is the work of post-publication empirical study, not white-paper-stage analysis.

15.4 C.4 Open Mathematical Questions

Three questions are deliberately left open and are candidates for future formalization.

First, the supermodularity proposition is stated for scalar information levels. Vector-valued generalization — where θ and ϕ each have multiple components corresponding to different aspects of model and embodiment understanding — preserves the supermodularity property under reasonable conditions but requires more careful proof. We sketch the argument in the GitHub companion repository.

Second, the Master Equation treats the five factors as independent multiplicative inputs. In practice, the factors are weakly correlated (e.g., higher N_{VLA} tends to drive higher $|\Sigma|$ as more VLAs contribute skills). Formal treatment of these correlations would refine the equation but is unlikely to overturn its qualitative implications.

Third, the constraint set is stated as deterministic. A stochastic version — in which each constraint can be violated with small probability without immediately collapsing T_{cert} — would more accurately capture real-world governance dynamics, and is the more defensible formulation. The deterministic form is retained in the body only as an expository simplification: because T_{cert} enters multiplicatively, a constraint violated with non-negligible probability still drives the *expected* value sharply down, so the qualitative policy implication — that the constraints are architectural preconditions rather than aspirational targets — is robust to the choice between the two formulations.

End of v1 draft.